



Gale Associates, Inc.

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December 12, 2017

Mr. Sean Mangano
Amherst-Pelham Finance Director
Amherst-Pelham Regional School District
170 Chestnut Street
Amherst, MA 01002

Re: 30% Schematic Design Submission
Roof Replacement and Associated Work at the
Amherst Regional Middle School
170 Chestnut Street
Amherst, MA 01002
Gale JN 833180 P1

Dear Mr. Mangano:

In accordance with our contract, Gale Associates Inc. (Gale) has prepared a brief schematic design submission for the Amherst-Pelham Regional School District (APRSD). This submission relates to the roof replacement and associated work at the Amherst Regional Middle School (School), located at 170 Chestnut Street in Amherst, Massachusetts. Within this submission, we summarized our evaluation of the existing conditions and outlined various recommendations and options for the replacement of the various roof areas and associated components at the School. Gale did not review the auditorium roofs and connecting corridor roofs as directed by APRSD. This report is augmented with photographic documentation and 30% schematic design drawings found in the Appendix of this letter.

As you are aware, the intent of this project is to remove and replace the existing roof components to provide a watertight roof system. Several roof systems were considered; however, it is Gale's understanding that APRSD would prefer a thermoplastic single-ply roof system with a 20-30-year system warranty. As such, Gale performed a visual evaluation of the existing roof, steel framed structure, and associated components on the designated roof areas to determine the condition of the roof system. Roof test cuts were performed by Titan Roofing, Inc. of Springfield, MA for verification of the existing roof system, to observe as-built construction of the roof assembly, securement methods, and conditions of the various roof system components. However, these test cuts allowed Gale to visually observe less than one (1) percent of the roof system, and therefore, unforeseen conditions may exist at this time and some additional investigation may be required as the design progresses. Locations of test cuts are shown on the attached roof area plan.

CELEBRATING 50 YEARS



It is Gale's understanding that the existing roof system was installed circa 1994 and the original school building was built in the 1960s, which included a gravel surface built-up roof. Based on our understanding of the existing building conditions and reports from APRSD staff, there are multiple locations where water infiltration has been observed in the past. It was reported by APRSD staff that approximately 75 repairs have been performed to mitigate moisture infiltration throughout the various roof areas in the form of silicone sealant. Gale observed several of these repairs during the overall review of the roof system.

Gale performed a leak audit of the interior spaces below the various roof areas in an attempt to determine the cause of the moisture infiltration. Gale also took this opportunity to briefly observe any potentially damaged/corroded sections of the structural roof deck and structural framing members as a result of the moisture infiltration. It does not appear the steel deck has been corroded from moisture infiltration at the limited locations which were observed, however, in locations where stained ceiling tiles were present, small holes in the steel deck were noted. These holes appear to be a result of failed or missed puddle welds and were often noted throughout the steel deck.

APRSD also requested that Gale review the existing roof related structural components to determine the feasibility of installing a photovoltaic (PV) system on the various roof areas. Gale performed a preliminary structural analysis based on the original roof framing members observed during the interior leak audit, as well as the framing members shown in the existing structural drawings generated by Alderman & MacNeish Architects, August 1967, provided by APRSD. Based on our preliminary analysis, it appears the existing structural framing is made up of steel I-beams, steel bar joists, and a steel roof deck. After reviewing the original drawings and a comparison to current code requirements, it is Gale's opinion that the structural steel roof decks and framing members in place do not provide sufficient support for the additional weight of a typical PV system. Preliminary calculations indicate that the capacity of the existing roof structure is already limited, and an additional dead load increase by a PV system will require significant augmentation of the existing roof structure. However, with the consultation of a PV designer and the associated augmentation of the existing roof structure by a licensed structural engineer, a PV system may be installed appropriately onto the existing structure but will require additional engineering to review the potential structural scope, which will be dependent on the type, configuration, and layout of a PV system supported over the roof. Due to the configuration of the building, it is anticipated that the existing roof deck would require complete removal in the areas where the PV system is to be installed to accommodate augmenting the existing steel joists to distribute the loads. Alternatively, new steel stub columns could be continued from the roof to the existing column system and would require a review of the foundation system which is beyond the limits of this review. Either option will likely not be cost effective or provide a payback return if a PV is considered on the roof system.

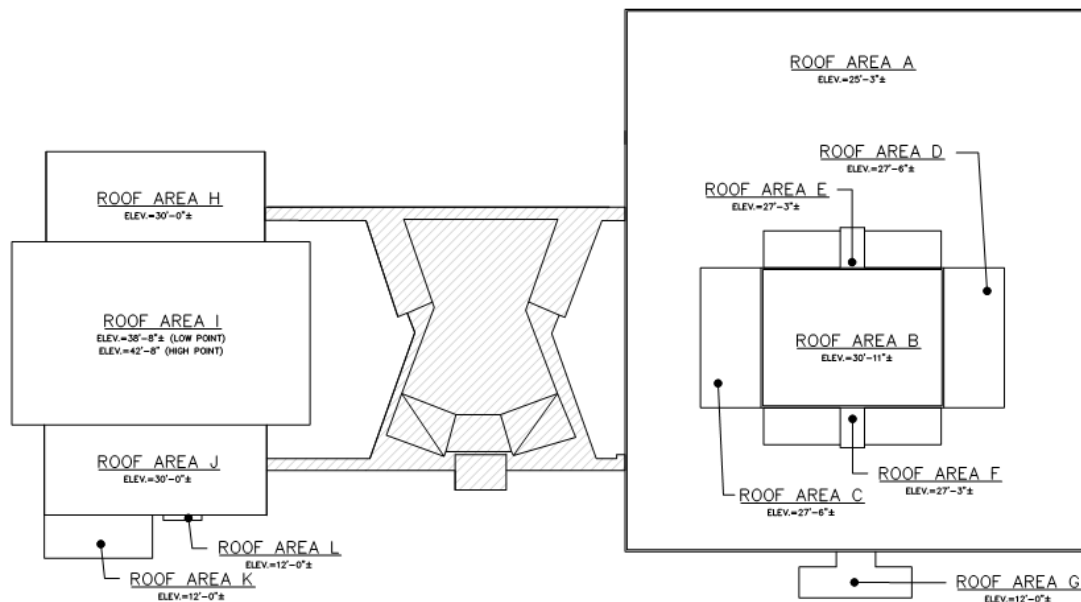


Figure 1 – Overall Roof Area Plan

It is Gale's understanding that the 9th Edition of the Massachusetts State Building Code (MSBC) will take full effect as of January 1, 2018. As such, it is anticipated that the permitting phase of this project will happen under the 9th Edition of the MSBC.

Gale's code review analysis can be found in the Appendix of this letter. Please refer to this section for the various codes that are expected to apply to this particular roof replacement project. As part of Gale's code review, wind uplift calculations were performed to determine the uplift force in pounds per square foot that will be required of the new roof system per ASCE 7-10. Gale also performed a drainage capacity analysis for the existing roof drains and it appears that each existing roof drain has sufficient diameter to provide drainage required to meet 248 CMR 10 regulations. The slope of the horizontal drain leaders within the building is unclear at this time. However, it is assumed that most leader pipes are installed at a 1/4" per foot slope. Using this assumption, each drain leader pipe appears to meet the requirements of 248 CMR 10 regulations with the exception of four (4) different sections of pipe. These sections of pipe service several different sections of Roof Area A. The slopes of these pipes should be confirmed where possible, to determine if larger diameter pipes should be installed to meet 248 CMR 10 regulations.

Recommended Scope of Work:

The following recommendations are based on the conditions observed in the field and the engineering parameters reviewed. The recommendations list the suggested scope of work to properly replace the roof system and reduce future maintenance costs. Not every recommendation has been listed here. The recommendations listed below are those which Gale believes will have



the greatest impact on cost and those which may have potentially been overlooked during the initial budget analysis performed by APRSD. The recommendations are as follows:

1. Remove the existing adhered thermoplastic roof membrane, insulation, loose-laid vapor retarder (where present), and associated components down to the existing roof deck, and install an adhered thermoplastic roof membrane, adhered high density coverboard, mechanically attached polyisocyanurate insulation, vapor retarder, and baseboard. The thickness of the proposed new roof assembly is anticipated to increase approximately 3-inches to meet the minimum R-Value (R-30) requirements set by the 9th Edition of the MSBC.
2. The throughwall flashing at the roof to wall transition on Roof Area K is currently low, and due to the anticipated new tapered insulation thickness, the existing roof flashing height at the base of the wall would be much less than the 8-inch recommended minimum industry design standard. Therefore, the existing throughwall flashing and brick masonry would require removal and replacement. The throughwall flashings on Roof Areas E, F, H, and J also appear that they will be low once the new roof system is installed. However, the flashing heights in these locations will not greatly differ from those which are existing. Therefore, unless otherwise requested, the existing throughwall flashings can remain in place at the rising walls on these Roof Areas. New throughwall flashing will also be required at four (4) locations at the parapet to rising wall transition on Roof Areas H and J. Similar conditions exist at four (4) small rising wall locations on Roof Area A where new throughwall flashings will also be required.
3. Gale was informed by APRSD that the brick masonry chimney on Roof Area A had been recently rebuilt and should remain in place as part of these renovations. However, due to the anticipated new tapered insulation thickness, the existing roof flashing height at the base of the chimney would be less than the 8-inch recommended minimum industry design standard. Therefore, the existing throughwall flashing and brick masonry would require removal and replacement.
4. On Roof Area J, the existing brick masonry chimney appears to have failed mortar joints, step cracking, and is in overall poor condition. Since APRSD reported that the chimney is no longer in use, Gale recommends removing the chimney down below the roof deck and infilling the roof deck in this location. Further review of the chimney's construction below the deck and written confirmation from the APRSD indicating that the chimney is no longer in use will be required. Please note that the existing chimney currently has lightning protection mounted to the chimney cap. If the chimney is removed, the lightning protection will likely need to be re-located or removed. The assistance of a lightning protection consultant will be required should APRSD decide to upgrade their lightning protection systems. Should APRSD consider keeping the chimney, Gale recommends repointing 100% of the mortar joints and rebuilding more than 50% of the outer wythe of the brick. The cost to repair the chimney would be approximately \$20,000 - \$30,000.



5. The brick masonry rising wall on Roof Area J appears to be experiencing step cracking in an isolated location, likely due to differential movement or settlement, as the crack is located just above the roof edge. It is recommended that this section of brick masonry be rebuilt to reduce the potential of further cracking or failure of the brick masonry veneer and moisture infiltration. It is also recommended that a masonry expansion joint be sawcut into the wall near the location of this cracking to allow for differential movement within the wall system.
6. The masonry vertical expansion joints located at the brick masonry rising walls on Roof Areas H and J appear to have missing or failed mortar joints. It is recommended that these expansion joints be replaced with sealant under the scope of this project to reduce for the potential of moisture infiltration into the new roof system or into the building. These rising walls both contain a louver with failed perimeter sealants. These sealants should be removed and replaced.
7. The majority of the existing cast iron drain bowls currently have copper sleeve inserts installed within the bowl and into the leader pipe. It appears these sleeves may have been installed with the new drains during the previous roof replacement as a substitute to extending the cast iron drain leaders. However, these configurations do not meet code and it is likely that the drain leaders will need to be extended with a no-hub connection below the roof deck. This will require the contractor to have interior access to the underside of each drain bowl. In addition, it is recommended that the existing roof drain bowls, clamping rings, hardware, and dome strainers be replaced. Install drain bowl collar extensions to accommodate the required minimum thickness of insulation. Provide roof crickets between roof drains with drain sumps to promote drainage and manage water.
8. It is recommended that any sections of the structural steel deck which appear to be loose or poorly connected to the structural framing be re-secured as part of this roof replacement project. Since the potential extent of unsecured decking is unknown at this time, it is recommended that the perimeters and corners of the roof be re-secured as part of the contract documents, and any missing fasteners/welds in the field of the roof will likely be covered by a unit cost, should this unforeseen condition be discovered during construction.
9. As the overall thickness of the insulation and roof assembly are increased, the height of the perimeter wood blocking and sheet metal edge fascia will also need to be increased. Gale recommends installing additional wood blocking to match the maximum height of insulation and installing new sheet metal edge fascia to terminate the perimeter of the roof. Note that the new sheet metal fascia may incorporate a 2 or 3-piece fascia system, which will change the aesthetics of the existing edge profile.



10. Remove and replace the existing roof access hatches located on Roof Areas A, H and J, and remove and infill two (2) abandoned roof access hatches that were reported on Roof Area B.
11. Remove and replace the roof access ladders on Roof Areas E and H in order to meet code requirements with the new thickness of the proposed roof assembly. The ladder on Roof Area H currently has a weather antenna device mounted to it. If APRSD requires that this device stay, then it must be relocated.
12. Perform temporary disconnection/reconnection as required to raise the existing mechanical units set on curbs to properly flash and/or provide an 8-inch minimum flashing height to achieve the recommended industry design standards.
13. Provide perimeter guardrails along roof access points and around mechanical units within 15-feet of any roof edge.

The recommended construction budget to complete the items presented above is suggested to be approximately \$2,471,000.00, which is approximately \$23/SF. This budget also includes a fifteen (15) percent contingency in the event that unforeseen conditions are encountered as part of the design or construction services. Note that these estimates are preliminary as the actual construction schedule, scope of work, and construction details have not been fully defined. This estimate is intended to provide APRSD an indication as to what construction costs may be given the existing conditions and Gale's recommendations. Gale does not recommend using this estimate for sensitive budgeting. Refer to the list below for each budget mentioned in the recommended scope of work. Please note, each of these costs include the anticipated costs for general conditions, mobilization, bonds, and insurance. Refer to the cost estimate located in the Appendix of this report for a more detailed summary.

Remove and Replace Roof System.....	\$1,837,670.00
Remove and Replace Designated Throughwall Flashings	\$19,570.00
Remove and Replace Roof Hatches and Ladders / Infill Hatches on Roof Area B	\$19,315.00
Rebuild Step Cracked Area / Remove and Replace Masonry Control Joints / Misc. Masonry.....	\$7,200.00
Remove and Replace Drains / Extend Leader Pipes / Extend Vent Pipes	\$99,180.00
Add Wood Blocking Around Perimeters	\$11,400.00
Re-secure Metal Deck at Corners and Perimeters.....	\$20,380.00
Provide Perimeter Guardrails Around Mechanical Units Within 15-feet of an Edge.....	\$4,000.00
General Conditions: Dumpsters / Cleanup / Temp. Protection / Etc.....	\$129,600.00
 Subtotal.....	 \$2,148,320.00
15% Contingency	\$322,248.00
Total Recommended Construction Budget	\$2,471,000.00

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30% Schematic Design Submission
Roof Replacement and Associated Work at the
Amherst Regional Middle School
December 12, 2017
Page 7



Please be reminded that these are preliminary estimates and are subject to change based on the bid climate, material and worker availability, and potential site restrictions with working at the School. The estimates are based on working during normal business hours, and do not include soft costs associated with working with APRSD, interior renovations beyond that which is normal for roof renovations, APRSD's administration of the project, or additional engineering costs should additional design options be considered.

Once APRSD has had the opportunity to review and select a proposed roof replacement option, and authorizes the project to proceed to the next phase, Gale will initiate the development of the Construction Documents based on the approved system. Please advise if you would like to have a conference call, or meet to discuss the proposed options.

Thank you for your attention. Should you have any questions or comments, please do not hesitate to contact us at this office.

Best regards,

GALE ASSOCIATES, INC.

Cole T. McCarthy/pm

Cole T. McCarthy, EIT
Building Envelope Design and Consulting Group

Christopher Musorofiti/pm

Christopher Musorofiti, RRC
Associate

CTM/pm
Attachments

cc: John Coty – APRSD
Samuel Moores – Gale
Sarah Frankenberger – Gale

Photographic Documentation:



Photo 1 – Partial overall view of the Middle School Roof Areas (A-L).



Photo 2 – Partial view of the recently rebuilt brick masonry chimney on Roof Area A.



Photo 3 – Overall view of the abandoned chimney on Roof Area J which appears to be in poor condition and is reportedly not used anymore.



Photo 4 – View of the roof to wall transition that will require new throughwall flashing at Roof Area K.



Photo 5 – View of typical burned holes within the steel deck which may have been caused during the misalignment of puddle-welds connecting steel deck and joist.



Photo 6 – View of one (1) masonry control joint at the rising wall above Roof Area J. These joints typically are in failed condition.



Photo 7 – View of the roof hatch on Roof Area H which will need to be replaced; similar to the ones on Roof Areas A and J.



Photo 8 – View of the roof access ladder on Roof Area H which will need to be replaced.



Photo 9 – Rooftop mechanical units will need to be disconnected, raised, and reconnected.



Photo 10 – Roof drain bowls appear to have copper sleeve inserts.



Appendix A

Preliminary Cost Estimate

Date: December 12, 2017	PRELIMINARY BUDGET ESTIMATE				SUMMARY SHEET				
Project Title: Amherst Regional Middle School	Gale JN: 833180								
Owner: Amherst-Pelham Regional School District	Estimated by: Gale Associates, Inc.								
	Status: Schematic Design for Study Submission								
	Engineering Estimate								
		Subtotal 1	General Conditions	Mobilization	Bonds and Insurance	Overhead and Profit	Subtotal 2	Total	MAIN SCOPE OF WORK
Description			(Subtotal x 7.5%)	(Subtotal x 5%)	(Subtotal x 3%)	(Subtotal X 15%)	(Including General Conditions, and Overhead and Profit)	(Subtotal 2 with 15% Contingency)	
Division 2 - General Conditions									
Temporary Interior and Overhead Protection, Disposal, Crane - Classroom Wing		\$57,000	\$4,275	\$2,850	\$1,710	\$8,550	\$74,385	\$86,000	
Temporary Interior and Overhead Protection, Disposal, Crane - Gym/Pool Wing		\$42,300	\$3,173	\$2,115	\$1,269	\$6,345	\$55,202	\$63,500	
Division 4 - Masonry									
Associated Masonry Work - Classroom Wing		\$4,200	\$315	\$210	\$126	\$630	\$5,481	\$6,500	
Associated Masonry Work - Gym/Pool Wing		\$16,315	\$1,224	\$816	\$489	\$2,447	\$21,291	\$24,500	
Division 7 - Roof Replacement									
Roof Replacement and Associated Work - Classroom Wing		\$1,041,800	\$78,135	\$52,090	\$31,254	\$156,270	\$1,359,549	\$1,563,500	
Roof Replacement and Associated Work - Gym/Pool Wing		\$408,600	\$30,645	\$20,430	\$12,258	\$61,290	\$533,223	\$613,500	
Division 22 - Plumbing									
Associated Plumbing Work - Classroom Wing		\$58,000	\$4,350	\$2,900	\$1,740	\$8,700	\$75,690	\$87,500	
Associated Plumbing Work - Gym/Pool Wing		\$18,000	\$1,350	\$900	\$540	\$2,700	\$23,490	\$27,500	
		Subtotal - Classroom	\$1,161,000	\$87,075	\$58,050	\$34,830	\$174,150		
		Subtotal - Gym/Pool	\$485,215	\$36,391	\$24,261	\$14,556	\$72,782		
		Subtotal for Roof Replacement and Associated Work - Classroom					\$1,515,105	\$1,742,500	
		Subtotal for Roof Replacement and Associated Work - Gym/Pool					\$633,206	\$728,500	
		Subtotal for Roof Replacement and Associated Work					\$2,148,311	\$2,471,000	
*Anticipated Construction Budget :								\$2,471,000.00	
Note: The construction cost estimates have been compiled from various sources and may be greater or less than actual construction costs. The estimated costs are considered preliminary budget figures only. The actual costs of construction could vary significantly based on the final selection of the scope of work and duration of the project. These estimates are intended to give APRSD an indication as to what construction costs may be, given the existing conditions and Gale's recommendations. Gale does not recommend utilizing these estimates for sensitive budgeting.									

PRELIMINARY BUDGET ESTIMATE				
Division 2 - General Conditions				
Item Description	Quantity		Engineering Estimate	
	Number	Unit	Unit Cost	Total Cost
Interior Temporary Protection - Classroom		Square Feet		\$15,000.00
Interior Temporary Protection - Gym/Pool		Square Feet		\$6,132.00
Overhead Protection at the 2 Main Entrances - Classroom		Unit		\$2,000.00
Overhead Protection at the 1 Main Entrances - Gym/Pool		Unit		\$1,000.00
Final Cleaning - Classroom		Square Feet		\$15,000.00
Final Cleaning - Gym/Pool		Square Feet		\$6,132.00
Disposal Charges - Classroom		Unit		\$45,000.00
Disposal Charges - Gym/Pool		Unit		\$19,000.00
Crane/Lull - Both		Unit		\$6,000.00
Demo Chimney on Roof J - Gym/Pool		E.A.		\$4,000.00
Materials and Labor (Classroom) Sub Total:				\$57,000.00
Materials and Labor (Gym/Pool) Sub Total:				\$42,264.00
Materials and Labor Sub Total:				\$99,264.00

PRELIMINARY BUDGET ESTIMATE				
Division 4 - Masonry				
Item Description	Quantity		Engineering Estimate	
	Number	Unit	Unit Cost	Total Cost
Roof Area A - Brick Masonry Chimney				
Remove and Replace Brick Masonry, Install New Throughwall Flashing (Chimney)		L.F.		\$3,000.00
Remove and Replace Brick Masonry, Install New Throughwall Flashing Roof Area A		L.F.		\$1,200.00
Remove and Replace Brick Masonry, Install New Throughwall Flashing (Roof Area K)		L.F.		\$9,000.00
Remove and Replace Brick Masonry, Install New Stepped Throughwall Flashing Where Parapet Meets Rising Wall (4 locations) (Roof Areas H and J)		L.F.		\$1,800.00
Brick Repointing (Unit Price)		S.F.		\$1,200.00
Brick Replacement (Unit Price)		E.A.		\$650.00
Rebuild Step Cracked Brick		L.F.		\$1,500.00
Remove and Replace Expansion Joints		L.F.		\$680.00
Sawcut and Install New Expansion Joint		L.F.		\$765.00
Remove and Replace Louver Perimeter Sealants		L.F.		\$720.00
Materials and Labor (Classroom) Sub Total:				\$4,200.00
Materials and Labor (Gym/Pool) Sub Total:				\$16,315.00
Materials and Labor Sub Total:				\$20,515.00

PRELIMINARY PHASED COST ESTIMATE - Amherst Regional Middle School				
Date: 12/12/2017	Gale Associates, Inc.			
Activity: Roofing and Associated Work	Gale JN: 833180			
Location: Amherst Regional Middle School	Estimated by: Gale Associates, Inc.			
Project Title: Amherst Regional Middle School				
Item Description		Quantity		Engineering Estimate
		Number	Unit	Total
ROOF AREA A				
Thermoplastic Single-Ply Roof Cover			S.F.	\$107,280.00
Coverboard (High Density Iso)			S.F.	\$41,720.00
Tapered polyisocyanurate Insulation			S.F.	\$298,000.00
Baseboard			S.F.	\$59,600.00
Vapor Barrier - self-adhered			S.F.	\$59,600.00
Tapered Edge Strip			S.F.	\$935.00
Roof Membrane Flashings			L.F.	\$3,700.00
Perimeter Metal Flashings (Pre-Fab/ Engineered)			L.F.	\$47,600.00
Expansion Joint - Roof to Roof			L.F.	\$5,025.00
Wood Blocking (2x6 at roof edge)			B.F.	\$3,400.00
Hook Strips			L.F.	\$8,500.00
Plywood Sheathing			L.F.	\$418.75
Termination Bars			L.F.	\$502.50
Walkway Pads			L.F.	\$7,500.00
Paint Antenna Base			E.A.	\$500.00
Replace Roof Hatches and Provide Guardrails			E.A.	\$3,500.00
Provide Perimeter Guardrails			E.A.	\$3,000.00
Screws/ Resecure Metal Decking (Perimeters and Corners)			E.A.	\$8,140.00
Metal Deck Infill (Unit Price)			S.F.	\$400.00
Screws/ Resecure Metal Decking (Unit Price)			E.A.	\$462.50
Repair Metal Decking (Surficial Rust) (Unit Price)			S.F.	\$120.00
Standard Roofing Sub-Total				\$660,000.00
Labor Estimate				
Total Roof Area	59,600	Square Feet		
Expected Daily Production per Crew	4,000	SqFT per Crew Day		
Expected Crew Size	12	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	8			
Roofing Labor Sub-Total				\$150,400.00
Total Roofing - Roof Area A				\$810,400.00
ROOF AREA B				
Thermoplastic Single-Ply Roof Cover			S.F.	\$14,148.00
Coverboard (High Density Iso)			S.F.	\$5,502.00
Tapered polyisocyanurate Insulation			S.F.	\$39,300.00
Vapor Barrier - self-adhered			S.F.	\$7,860.00
Baseboard			S.F.	\$7,860.00
Tapered Edge Strip			S.F.	\$198.00
Roof Membrane Flashings			L.F.	\$720.00
Perimeter Metal Flashings (Pre-Fab/Engineered)			L.F.	\$10,080.00
Wood Blocking (2x6 at roof edge)			B.F.	\$720.00
Hook Strips			L.F.	\$1,800.00
Remove Roof Hatches (2)			E.A.	\$1,000.00
Screws/ Resecure Metal Decking (Perimeters and Corners)			E.A.	\$2,035.00
Metal Deck Infill (Unit Price)			S.F.	\$300.00
Screws/ Resecure Metal Decking (Unit Price)			E.A.	\$92.50
Repair Metal Decking (Surficial Rust) (Unit Price)			S.F.	\$90.00
Standard Roofing Sub-Total				\$91,800.00
Labor Estimate				
Total Roof Area	7,860	Square Feet		
Expected Daily Production per Crew	4,000	SqFT per Crew Day		
Expected Crew Size	10	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	1			
Roofing Labor Sub-Total				\$16,800.00

Total Roofing - Roof Area B				\$108,600.00
ROOF AREA C				
Thermoplastic Single-Ply Roof Cover		S.F.		\$4,680.00
Coverboard (High Density Iso)		S.F.		\$1,820.00
Flat Stock polyisocyanurate Insulation		S.F.		\$13,000.00
Vapor Barrier - self-adhered		S.F.		\$2,600.00
Baseboard		S.F.		\$2,600.00
Tapered Edge Strip		S.F.		\$137.50
Roof Membrane Flashings		L.F.		\$700.00
Perimeter Metal Flashings (Pre-Fab/Engineered)		L.F.		\$7,000.00
Wood Blocking (2x6 at roof edge)		B.F.		\$500.00
Hook Strips		L.F.		\$750.00
Plywood Sheathing		L.F.		\$218.75
Termination Bars		L.F.		\$112.50
Walkway Pads		L.F.		\$1,000.00
Metal Deck Infill (Unit Price)		S.F.		\$50.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$37.00
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$15.00
Standard Roofing Sub-Total				\$35,300.00
Labor Estimate				
Total Roof Area	2,600	Square Feet		
Expected Daily Production per Crew	2,600	SqFT per Crew Day		
Expected Crew Size	6	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	2			
Roofing Labor Sub-Total				\$6,400.00
Total Roofing - Roof Area C				\$41,700.00
ROOF AREA D				
Thermoplastic Single-Ply Roof Cover		S.F.		\$4,680.00
Coverboard (High Density Iso)		S.F.		\$1,820.00
Flat Stock polyisocyanurate Insulation		S.F.		\$13,000.00
Vapor Barrier - self-adhered		S.F.		\$2,600.00
Baseboard		S.F.		\$2,600.00
Tapered Edge Strip		S.F.		\$137.50
Roof Membrane Flashings		L.F.		\$700.00
Perimeter Metal Flashings (Pre-Fab/ Engineered)		L.F.		\$7,000.00
Wood Blocking (2x6 at roof edge)		B.F.		\$500.00
Hook Strips		L.F.		\$750.00
Plywood Sheathing		L.F.		\$218.75
Termination Bars		L.F.		\$112.50
Walkway Pads		L.F.		\$1,000.00
Metal Deck Infill (Unit Price)		S.F.		\$50.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$37.00
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$15.00
Standard Roofing Sub-Total				\$35,300.00
Labor Estimate				
Total Roof Area	2,600	Square Feet		
Expected Daily Production per Crew	2,600	SqFT per Crew Day		
Expected Crew Size	6	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	2			
Roofing Labor Sub-Total				\$6,400.00
Total Roofing - Roof Area D				\$41,700.00
ROOF AREA E				
Thermoplastic Single-Ply Roof Cover		S.F.		\$540.00
Coverboard (High Density Iso)		S.F.		\$210.00
Flat Stock polyisocyanurate Insulation		S.F.		\$1,500.00
Vapor Barrier - self-adhered		S.F.		\$300.00
Baseboard		S.F.		\$300.00
Tapered Edge Strip		S.F.		\$41.25
Roof Membrane Flashings		L.F.		\$150.00
Perimeter Metal Flashings (Pre-Fab/Engineered)		L.F.		\$2,100.00
Wood Blocking (2x6 at roof edge)		B.F.		\$150.00

Hook Strips		L.F.		\$325.00
Plywood Sheathing		L.F.		\$18.75
Termination Bars		L.F.		\$22.50
Walkway Pads		L.F.		\$250.00
Roof Access Ladder		E.A.		\$2,000.00
Metal Deck Infill (Unit Price)		S.F.		\$50.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$18.50
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$15.00
Standard Roofing Sub-Total				\$8,000.00
Labor Estimate				
Total Roof Area	300	Square Feet		
Expected Daily Production per Crew	300	SqFT per Crew Day		
Expected Crew Size	2	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	1			
Roofing Labor Sub-Total				\$2,400.00
Total Roofing - Roof Area E				\$10,400.00
ROOF AREA F				
Thermoplastic Single-Ply Roof Cover		S.F.		\$540.00
Coverboard (High Density Iso)		S.F.		\$210.00
Flat Stock polyisocyanurate Insulation		S.F.		\$1,500.00
Vapor Barrier - self-adhered		S.F.		\$300.00
Tapered Edge Strip		S.F.		\$41.25
Baseboard		S.F.		\$300.00
Roof Membrane Flashings		L.F.		\$150.00
Perimeter Metal Flashings (Pre-Fab/ Engineered)		L.F.		\$2,100.00
Wood Blocking (2x6 at roof edge)		B.F.		\$150.00
Hook Strips		L.F.		\$325.00
Plywood Sheathing		L.F.		\$18.75
Termination Bars		L.F.		\$22.50
Walkway Pads		L.F.		\$250.00
Metal Deck Infill (Unit Price)		S.F.		\$50.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$18.50
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$15.00
Standard Roofing Sub-Total				\$6,000.00
Labor Estimate				
Total Roof Area	300	Square Feet		
Expected Daily Production per Crew	300	SqFT per Crew Day		
Expected Crew Size	2	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	1			
Roofing Labor Sub-Total				\$2,400.00
Total Roofing - Roof Area F				\$8,400.00
ROOF AREA G				
Thermoplastic Single-Ply Roof Cover		S.F.		\$2,340.00
Coverboard (High Density Iso)		S.F.		\$910.00
Tapered polyisocyanurate Insulation		S.F.		\$5,200.00
Vapor Barrier - self-adhered		S.F.		\$1,300.00
Tapered Edge Strip		S.F.		\$96.25
Roof Membrane Flashings		L.F.		\$350.00
Perimeter Metal Flashings (Pre-Fab/ Engineered)		L.F.		\$4,900.00
Wood Blocking (2x6 at roof edge)		B.F.		\$350.00
Hook Strips		L.F.		\$775.00
Plywood Sheathing		L.F.		\$31.25
Termination Bars		L.F.		\$37.50
Concrete Deck Repair (Unit Price)		S.F.		\$250.00
Standard Roofing Sub-Total				\$16,600.00
Labor Estimate				
Total Roof Area	1,300	Square Feet		
Expected Daily Production per Crew	1,300	SqFT per Crew Day		
Expected Crew Size	4	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	1			
Roofing Labor Sub-Total				\$4,000.00

Total Roofing - Roof Area G				\$20,600.00
ROOF AREA H				
Thermoplastic Single-Ply Roof Cover		S.F.		\$11,160.00
Coverboard (High Density Iso)		S.F.		\$4,340.00
Tapered polyisocyanurate Insulation		S.F.		\$31,000.00
Baseboard		S.F.		\$6,200.00
Vapor Barrier - self-adhered		S.F.		\$6,200.00
Tapered Edge Strip		S.F.		\$192.50
Roof Membrane Flashings		L.F.		\$700.00
Perimeter Metal Flashings (Pre-Fab/ Engineered)		L.F.		\$9,800.00
Wood Blocking (2x6 at roof edge)		B.F.		\$700.00
Hook Strips		L.F.		\$1,125.00
Plywood Sheathing		L.F.		\$156.25
Termination Bars		L.F.		\$187.50
Walkway Pads		L.F.		\$500.00
Roof Access Ladder		E.A.		\$2,000.00
Replace Roof Hatches with Guardrails		E.A.		\$3,500.00
Screws/ Resecure Metal Decking (Perimeters and Corners)		E.A.		\$1,239.50
Metal Deck Infill (Unit Price)		S.F.		\$50.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$37.00
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$15.00
Standard Roofing Sub-Total				\$79,200.00
Labor Estimate				
Total Roof Area	6,200	Square Feet		
Expected Daily Production per Crew	3,100	SqFT per Crew Day		
Expected Crew Size	8	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	2			
Roofing Labor Sub-Total				\$14,400.00
Total Roofing - Roof Area H				\$93,600.00
ROOF AREA I				
Thermoplastic Single-Ply Roof Cover		S.F.		\$30,150.00
Coverboard (High Density Iso)		S.F.		\$11,725.00
Flat Stock polyisocyanurate Insulation		S.F.		\$67,000.00
Baseboard		S.F.		\$16,750.00
Vapor Barrier - self-adhered		S.F.		\$16,750.00
Tapered Edge Strip		S.F.		\$297.00
Roof Membrane Flashings		L.F.		\$1,080.00
Perimeter Metal Flashings (Pre-Fab/ Engineered)		L.F.		\$15,120.00
Wood Blocking (2x6 at roof edge)		B.F.		\$1,080.00
Hook Strips		L.F.		\$2,700.00
Walkway Pads		L.F.		\$1,000.00
Screws/ Resecure Metal Decking (Perimeters and Corners)		E.A.		\$2,960.00
Metal Deck Infill (Unit Price)		S.F.		\$100.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$138.75
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$30.00
Standard Roofing Sub-Total				\$166,900.00
Labor Estimate				
Total Roof Area	16,750	Square Feet		
Expected Daily Production per Crew	6,000	SqFT per Crew Day		
Expected Crew Size	8	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	1			
Roofing Labor Sub-Total				\$20,000.00
Total Roofing - Roof Area I				\$186,900.00
ROOF AREA J				
Thermoplastic Single-Ply Roof Cover		S.F.		\$11,160.00
Coverboard (High Density Iso)		S.F.		\$4,340.00
Tapered polyisocyanurate Insulation		S.F.		\$31,000.00
Baseboard		S.F.		\$6,200.00
Vapor Barrier - self-adhered		S.F.		\$6,200.00
Tapered Edge Strip		S.F.		\$192.50

Roof Membrane Flashings		L.F.		\$700.00
Perimeter Metal Flashings (Pre-Fab/Engineered)		L.F.		\$9,800.00
Wood Blocking (2x6 at roof edge)		B.F.		\$700.00
Hook Strips		L.F.		\$1,125.00
Plywood Sheathing		L.F.		\$156.25
Termination Bars		L.F.		\$187.50
Walkway Pads		L.F.		\$1,000.00
Replace Roof Hatches with Guardrails		E.A.		\$3,500.00
Screws/ Resecure Metal Decking (Perimeters and Corners)		E.A.		\$1,239.50
Metal Deck Infill (Unit Price)		S.F.		\$50.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$37.00
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$15.00
Standard Roofing Sub-Total				\$77,700.00
Labor Estimate				
Total Roof Area	6,200	Square Feet		
Expected Daily Production per Crew	3,100	SqFT per Crew Day		
Expected Crew Size	8	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	3			
Roofing Labor Sub-Total				\$15,200.00
Total Roofing - Roof Area J				\$92,900.00
ROOF AREA K				
Thermoplastic Single-Ply Roof Cover		S.F.		\$2,610.00
Coverboard (High Density Iso)		S.F.		\$1,015.00
Tapered polyisocyanurate Insulation (1/8" / ft)		S.F.		\$7,250.00
Baseboard		S.F.		\$1,450.00
Vapor Barrier - self-adhered		S.F.		\$1,450.00
Tapered Edge Strip		S.F.		\$96.25
Roof Membrane Flashings		L.F.		\$350.00
Perimeter Metal Flashings (Pre-Fab/ Engineered)		L.F.		\$4,900.00
Wood Blocking (2x6 at roof edge)		B.F.		\$350.00
Hook Strips		L.F.		\$575.00
Plywood Sheathing		L.F.		\$75.00
Termination Bars		L.F.		\$90.00
Expansion Joint - Roof to Wall		L.F.		\$2,100.00
Metal Deck Infill (Unit Price)		S.F.		\$50.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$37.00
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$15.00
Standard Roofing Sub-Total				\$22,500.00
Labor Estimate				
Total Roof Area	1,450	Square Feet		
Expected Daily Production per Crew	1,450	SqFT per Crew Day		
Expected Crew Size	6	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	2			
Roofing Labor Sub-Total				\$6,400.00
Total Roofing - Roof Area K				\$28,900.00
ROOF AREA L				
Thermoplastic Single-Ply Roof Cover		S.F.		\$234.00
Coverboard (High Density Iso)		S.F.		\$91.00
Tapered polyisocyanurate Insulation		S.F.		\$520.00
Baseboard		S.F.		\$130.00
Vapor Barrier - self-adhered		S.F.		\$130.00
Tapered Edge Strip		S.F.		\$33.00
Roof Membrane Flashings		L.F.		\$120.00
Perimeter Metal Flashings (Pre-Fab/ Engineered)		L.F.		\$1,680.00
Wood Blocking (2x6 at roof edge)		B.F.		\$120.00
Hook Strips		L.F.		\$200.00
Plywood Sheathing		L.F.		\$25.00
Termination Bars		L.F.		\$30.00
Scupper and Downspout		E.A.		\$500.00
Metal Deck Infill (Unit Price)		S.F.		\$20.00
Screws/ Resecure Metal Decking (Unit Price)		E.A.		\$9.25
Repair Metal Decking (Surficial Rust) (Unit Price)		S.F.		\$6.00

Standard Roofing Sub-Total				\$3,900.00
Labor Estimate				
Total Roof Area	130	Square Feet		
Expected Daily Production per Crew	130	SqFT per Crew Day		
Expected Crew Size	2	Men/Crew		
Cost Per Man Day	\$800.00	Dollars/Man Day		
Additional Work (Flashings)	1			
Roofing Labor Sub-Total				\$2,400.00
Total Roofing - Roof Area L				\$6,300.00
Total Roofing - Classroom Building				\$1,041,800.00
Total Roofing - Gym/Pool Building				\$408,600.00
Total Roofing				\$1,450,400.00

PRELIMINARY BUDGET ESTIMATE				
Division 22 - Plumbing				
Item Description	Quantity		Engineering Estimate	
	Number	Unit	Unit Cost	Total Cost
Roof Area A - Brick Masonry Chimney				
Roof Area A - Remove and Replace Roof Drains (Extend Leader Pipe, Bowl, new ring, strainer, and hardware, extension collar)		E.A.		\$40,000.00
Roof Area B - Remove and Replace Roof Drains (Extend Leader Pipe, Bowl, new ring, strainer, and hardware, extension collar)		E.A.		\$8,000.00
Roof Area G - Remove and Replace Roof Drains (Extend Leader Pipe, Bowl, new ring, strainer, and hardware, extension collar)		E.A.		\$4,000.00
Roof Area H - Remove and Replace Roof Drains (Extend Leader Pipe, Bowl, new ring, strainer, and hardware, extension collar)		E.A.		\$4,000.00
Roof Area I - Remove and Replace Roof Drains (Extend Leader Pipe, Bowl, new ring, strainer, and hardware, extension collar)		E.A.		\$4,000.00
Roof Area J - Remove and Replace Roof Drains (Extend Leader Pipe, Bowl, new ring, strainer, and hardware, extension collar)		E.A.		\$4,000.00
Roof Area K - Remove and Replace Roof Drains (Extend Leader Pipe, Bowl, new ring, strainer, and hardware, extension collar)		E.A.		\$4,000.00
Extend existing vent pipes (Roof Areas A-G)		E.A.		\$6,000.00
Extend existing vent pipes (Roof Areas H-L)		E.A.		\$2,000.00
Materials and Labor (Classroom) Sub Total:				\$58,000.00
Materials and Labor (Gym/Pool) Sub Total:				\$18,000.00
Materials and Labor Sub Total:				\$76,000.00



Appendix B

Code Review



**ROOF REPLACEMENT AND ASSOCIATED WORK
AT THE AMHERST REGIONAL MIDDLE SCHOOL
Amherst Regional School District
Amherst, MA**

GALE JN 833180

GENERAL DESIGN CONSIDERATIONS

This section relates to specific design considerations with respect to the removal and replacement of the roof systems and associated components that may be affected by the proposed installations at the Amherst-Pelham Regional Middle School (School).

BUILDING CODE ASSESSMENT

New construction, as well as repair and alteration of existing buildings in Massachusetts, is regulated by 780 CMR, the Massachusetts State Building Code (MSBC), and 521 CMR, the Architectural Access Board (AAB).

The 9th Edition of the MSBC will take full effect as of January 1, 2018. Therefore, it is anticipated that the permitting phase of this project will happen under the 9th Edition of the MSBC. The following code review is based upon the following:

- 2015 *International Building Code* (IBC),
- 2015 *International Existing Building Code* (IEBC),
- 2015 *International Energy Conservation Code* (IECC),
- 2015 *International Mechanical Code* (IMC),
- CMR 248 *Massachusetts Uniform State Plumbing Code*,
- Massachusetts Amendments.

Applicable portions of these pertinent codes are as follows:

780 CMR Massachusetts State Building Code

Under the 9th edition of the MSBC, *Chapter 34 Existing Building and Structures* refers to the *International Existing Building Code 2015* (IEBC) as modified by the Massachusetts Amendments. This section of the code applies when there are repairs, alterations, additions or a change in use to existing buildings and generally refers to other sections of 780 CMR for specific requirements. Alterations to existing buildings, including the removal and replacement of building elements with a continuation of the same use group (as is the case for the roof replacement at the PPB), are governed by *Chapter 3 – Provisions for All Compliance Methods*, and *Chapter 7, Alterations – Level 1*.

Chapter 3 (IEBC) – Provisions for All Compliance Methods

The 9th Edition Amendments include changes to Chapter 3 of the IEBC. The following requirements have been added and address structural requirements for unreinforced masonry parapets:

Section 302.7.1: Structural Requirements Pertaining to Roof Work, Seismic – This section requires that parapets of unreinforced masonry be analyzed for bracing (403.5 and 707.3.1) where the building is assigned to Seismic Design Category D, E or F. The School is in Seismic Design Category B and does not have parapets, therefore this section does not apply to this project.

Section 302.7.2: Structural Requirements Pertaining to Roof Work, Wind – This section requires that parapets of unreinforced masonry be analyzed for bracing (403.8 and 707.3.2) where the ultimate design wind speed is greater than 150 mph and the building is occupancy category IV per Table 1604.5 of the IBC. As the ultimate design wind speed for Amherst, MA is 125 mph (Table 1604.11), this section does not apply.

Chapter 7 (IEBC) – Alterations Level 1

Section 702: Building Elements and Materials – Generally, existing buildings must be modified in a manner that does not decrease safety. Existing items may remain, but all new materials to be used shall be compliant with code for new construction.

Section 703: Fire Protection – This section of the code defines requirements for fire resistant materials. Generally, for an existing building, the building layout and fire resistance may be presumed to have met the codes in effect at the time of construction per the IEBC. New work must conform to the building code for new construction, but upgrade of the existing building components to remain is not required. If during construction, existing fire resistant materials to remain are damaged or removed, they must be replaced to meet the requirements for new construction. The IBC, Chapter 15, Section 1505 - Fire Classification requirements were also reviewed.

Based on Gale's evaluation, the building construction throughout the School appears to be consistent. Based on the classifications in the MSBC, this facility houses classrooms, gymnasium, and an indoor pool and is considered to be Use Groups E (Educational), and A (Assembly) occupancy. Generally, Gale has found that the building is Type 2B construction. Therefore, the roof coverings shall be classified in accordance with the IBC when tested in accordance with ASTM E108.

Based on information provided in the MSBC, it appears that Class C fire rated roof assemblies, as tested and approved by Underwriter's Laboratories (U.L.) are acceptable for this project. Gale typically recommends that whenever possible, these types of structures have Class "A" roof systems designed and installed to provide better resistance to fire propagation from external sources on low slope roofs. Each of the low slope roof systems and components recommended will typically meet or exceed U.L. requirements, though the steep sloped stairwell will be detrimental as a Class C system.

Egress and Accessibility

Section 704: Means of Egress – Any alterations shall be done in a manner that maintains the level of protection of the means of egress. The project scope includes the removal and replacement of the existing roof assembly; therefore, Gale anticipates no modifications to egress components as part of this project.

Section 705: Accessibility – A facility that is altered shall comply with 521 CMR Massachusetts Architectural Access Board (MAAB) as well as provisions of the International Existing Building Code (IEBC). The IEBC requires where an alteration affects a space containing a primary function, then the

route to the space should be made accessible including restrooms and drinking fountains serving the primary space. The requirement for providing alterations is capped at 20% of the cost of the alteration. APRSD indicated that the School was in compliance with MAAB, so it is not anticipated that these types of renovations will be required for this project.

Reroofing Components

Section 706: Reroofing – This section of the code outlines requirements where reroofing is required, including drainage of low-sloped roofs, structural and construction activity load requirements, and flashing requirements.

Although reroofing shall not be required to meet ¼-inch per foot slope, positive slope to drains must be provided, and a ponding instability structural analysis is required to confirm that the roof structure possesses adequate stiffness to preclude progressive deflection in the event that the primary drain system becomes obstructed. Roof surfaces which have at least ¼-inch per foot slope are not considered to be susceptible to ponding instability, and as such, it will be the design intent to provide ¼-inch per foot slope to drain for the proposed roof renovations.

Structural Review

Section 707: Structural – This section of the code outlines requirements where alterations of work include replacement of equipment that is supported by the building or where a reroofing permit is required.

As part of the evaluation of the existing roofing system, Gale completed a code load review of the existing building's roof framing in accordance with the IBC.

707.2 Addition or replacement of roofing or replacement of equipment – This section of the code outlines requirements for vertical loading on structural elements from roof recoverers and replacements. The code states that if the proposed roof system increases the force in the supporting structural element by more than 5%, those structural elements shall be strengthened, supplemented, replaced, or altered to carry the increased load if the elements are found to have insufficient capacity. As such, dead and live roof gravity loads, using weights of the proposed materials versus original roof systems, were reviewed to verify load capacity. It appears that the proposed system will not exceed the five (5) percent threshold with respect to the original roof system. Note that the additional weight for the proposed photovoltaic system was not considered in the above reference dead load comparison.

707.3.1: Bracing for unreinforced masonry bearing wall parapets - Where a permit is issued for reroofing the entire roof diaphragm of a building assigned to Seismic Design Category D, E, or F that has unreinforced masonry parapets which exceed a 2.5 height-to-thickness ratio, the work shall include the installation of bracing to resist the reduced IBC level seismic forces unless an engineering evaluation demonstrates compliance of the existing parapet assembly. However, the School does not have any masonry wall parapets and therefore, this section of the code does not apply.

707.3.2: Roof diaphragms resisting wind loads in high-wind regions - Where a permit is issued for reroofing the entire roof diaphragm of a building located where the ultimate design wind speed (V_{ult}) is greater than 150 mph and the building is categorized as Risk Category IV, evaluation of the roof diaphragms and connections that are part of the main wind-force resisting system is required. Because the ultimate wind speed for the Town of Amherst is 125 mph, and the building appears to be classified as Risk Category III, this section of the code does not apply for this project.

In addition to Section 707 of the IEBC, Chapter 15 and Chapter 16 of the IBC provides structural requirements which were reviewed as part of this project.

Chapter 15: Roof Assemblies and Rooftop Structures – This section relates to wind resistance requirements for roof assemblies.

Section 1504.1 Wind resistance of roofs – Wind uplift requirements for low-slope roof areas should be performed in conformance with both the MSBC, which references the American Society of Civil Engineers (ASCE) design standard ASCE 7-10, and the Factory Mutual (FM) Global Method, as noted within Data Sheet 1-28, “Wind Design.” Increased attachment methods are typically provided at the perimeter and corners of each roof area replaced to withstand the increased uplift pressures for these specific areas. Refer to the Chapter 16 commentary for further structural requirements regarding wind uplift.

Chapter 16: Structural Design – This section states the requirements for structural loads on the roof, wall, and window systems, including dead, live, snow, wind, and seismic loads. A structural wind analysis has been performed, including wind uplift analysis of the roof areas, and is described below.

Section 1608 Snow Loads – Gale reviewed the snow load requirements for the low-slope roof areas, based on the structural drawings provided to Gale, in conformance with the MSBC, which references the American Society of Civil Engineers (ASCE) design standard ASCE 7-10, Chapter 7. Refer to the live load analysis below for a summary of the snow load analysis. The calculations are provided in the appendix sections of this report for additional information.

Wind Uplift (IBC Chapter 16, Section 1609 as referenced in MSBC Chapter 15, Section 1504.3) - Gale reviewed the wind uplift requirements of the typical low-slope roof areas in conformance with the MSBC, 9th Edition, which references the American Society of Civil Engineers (ASCE) design standard ASCE7-10, and the Factory Mutual (FM) Global Method, as noted within Data Sheet 1-28 Wind Loads to Roof Systems and Roof Deck Securement.

Our calculations indicate that the roof areas should be capable of withstanding the following minimum uplift pressures. The requirements for the roof system wind uplift resistance are:

Low-slope Roof systems:

- Roof Area I: -42.5 pounds per square foot in the field of the roof or FM Class 1-90.

Increased attachment methods will be provided at the perimeter and corners of each roof area replaced to withstand the increased uplift pressures for these specific areas. It will be the intent of the specifications to install the proposed roof system in accordance with the FM Class 1-90 uplift requirements. Refer to the calculation sheets in the appendix of this report for additional information.

Gravity Load Analysis

Description of Roof System/ Building

The accessible roof areas at the classroom and gym/pool building typically feature only one type of roofing systems. The original building drawings, dated August 24, 1967, indicates that the main roof framing consists of a 1-1/2- inch corrugated steel decking. The steel deck is supported by a steel subframe system comprised of parallel chord-steel joists that rest on wide-flange steel girders or steel beams along the building perimeter. The framing plan for the gym/pool building indicates a troughed roof type with structural slopes provided by a single-pitched steel truss and several long span steel joists. The size and length of the steel members vary throughout the roof framing plan of both building structures. Original drawings indicate that the steel deck sheets are typically secured to the subframe joist system via staggered welds along the supporting joists, trusses and beams. Steel and concrete columns provide vertical support for the horizontal beam and joist members.

Dead Loads:

The existing building plans from the year 1967 indicate that the original roof system for the classroom and gym/pool building consisted of tar and gravel built-up roofing and approximately 2-inches of polyisocyanurate insulation. In total the original roof system weighed approximately 7.0 pounds per square foot (psf). The existing roof system, which was observed on the roof areas during Gale's evaluation, appeared to be comprised of a thermoplastic membrane covering with approximately 3-inches of polyisocyanurate insulation, and a plastic vapor retarder. The existing and current roof systems for all roof areas weighs approximately 3.5 psf. For the purposes of the code required review, the roof system weights are in addition to the ventilation hoods and roof top units installed over portions of the roof, the structural roof components, and any items suspended from the roof structure including, but not limited to, electrical, mechanical, and plumbing components. The proposed replacement thermoplastic roof system, which incorporates continuous insulation above the roof deck, weighs approximately 5.5 psf. Based on the code required preliminary calculations the existing dead load from the roofing does not increase by more than 5.0% and additional structural analysis and review of the existing structural roof framing with compliance of current codes is not required at this time. Gale's preliminary calculations were performed with the assumption that all structural components below the steel deck will remain in place.

Live Loads:

The ground snow load value for Amherst, Massachusetts is 40.0 psf. Utilizing the roof snow load evaluation procedures of ASCE 7-10 as specified by the MSBC 9th Edition, the calculated minimum flat roof snow load for the facility is 35.0 psf. The original drawings indicate that the roof structure was designed to support a 30.0 psf roof live load. In Gale's experience, buildings designed prior to 1978 did not account for loading due to snow drift since it had not been adopted into the building code at that time. Since the original building was constructed approximately in 1967, Gale performed a snow drift analysis on the current structural framing components. Roof framing members below Roof Areas H, J and K were analyzed under the calculated drifting snow loads and found to beyond the original design capacity. Snow drift loads are based upon the length of the roof, and as such the volume of snow that has the potential to drift against a rising wall. ASCE 7-10 requires that the drift loads be applied in addition to the flat roof

snow loads and provides drift limitations based upon the geometry of the projection. At the majority of the Roof Areas H, J and K adjacent to rising walls, drifting snow has the potential to add the design drift load of 103.6 psf. However, the proposed roof system load does not trigger the 5.0% threshold and load analysis based on current building code standards are not required. Original drawings indicate that the design load for the existing steel decking and steel joists is 52.0 and 60.0 psf respectively. Therefore, it appears that the structural components have sufficient capacity to support the proposed roof system dead loads and a flat roof snow load of 35.0 psf. Though it is Gale's interpretation of the building code that upgrades to the existing decking system are not required. APRSD may wish to consider augmentation of the deck in these areas to support the additional drifting snow loads, or anticipate that some snow removal may be required when snow drifting heights exceed two (2) feet above the roof. According to the set of drawings by Tessier Architects, Inc. titled "Roof Replacement, Amherst-Pelham Regional Junior High School" (1994), provided by APRSD, it appears Roof Areas C, D, E, and F were added to potentially reduce snow drift accumulation in these areas. This indicates that other portions of the steel roof decks may be susceptible to excessive snow drift accumulation, such as those listed above.

Photovoltaic Systems:

At the time of this report, a specific PV system or a consultant for solar structures has not been selected by APRSD. Based on our preliminary calculations with current code requirements, the capacity of the existing roof system is very limited. Therefore, a significant amount of structural augmentation is to be considered within the roof framing to support ballasted PV units or a photovoltaic array on the roof structure. At this time, it is not anticipated that the rate return and schedule implications for the installation of a PV system on this building will be cost effective during the roof renovations. If APRSD wishes to continue with the PV installation, a more detailed engineering evaluation, load transfer down to the foundation and structural augmentation of the roof framing system is recommended, which is beyond the limits of this evaluation.

Energy Efficiency

Section 708: Energy Conservation – This section of the code dictates the requirements for thermal insulation for all exterior envelope components such as roofs, exterior walls, windows, and exterior doors. Gale reviewed the MSBC's Chapter 13, Energy Efficiency, which dictates the requirements for thermal insulation for all exterior envelope components, such as roofs and fenestrations, be designed in accordance with the IECC 2015, with the Massachusetts Amendments. Because only select areas of the building enclosure are being recommended for replacement, only the parts that are affected by the design and governed by the code should be brought to meet minimum energy requirements. These items may be limited to designated window(s) and roof renovations for this project. Below is a table highlighting the energy conservation requirements for the MSBC 9th edition:

Building System	MSBC 9 th Edition
Roof insulation entirely above the roof deck	R-30

Definition of Terms

R = Thermal Resistance ($h \times ft^2 \times ^\circ F / BTU$)

For roof renovations, it is Gale's recommendation to meet the above listed MSBC requirements which states that the roof insulation will be based on the R-30 thermal resistance. For calculation purposes, the design criteria utilize rigid polyisocyanurate board insulation having a long-term transmission rate R-value of 5 per inch.

Another design consideration relating to IBC Chapter 13 – Energy Efficiency which should be considered by a prudent professional during subsequent project phases include a vapor drive/condensation analysis. Prior to finalizing the roof assembly, the interior building humidity and subsequent vapor drive is recommended to be analyzed for dew point or condensation potential.

Additional Considerations:

In addition to Gale's review of the IEBC, several other code requirements were reviewed as they pertain to the building envelope. The following code requirements are referenced in the IBC and the proposed draft Massachusetts Amendments.

Means of Egress and Fall Protection

Chapter 10 (IBC): Section 1011.16 Ladders (International Mechanical Code (IMC) 2015, Chapter 3, Section 306.5) – This section addresses the requirements for safe means to access and egress the roof. The IBC 1011.16 requires permanent ladders to be installed for roof access, whereas 2015 IMC describes the requirements of construction for ladders. They include rung spacing of not more than 12-inches, not less than 18-inches between rails, rungs shall be capable of withstanding a 300-lb load, and shall be permanently connected to the floor and assembly above.

Chapter 10 (IBC): Section 1015 Guards (International Mechanical Code (IMC) 2015, Chapter 3, Section 304.11) – This section of the code dictates that where mechanical equipment (fans/vents) and roof access/walkways are within 10-feet of the roof edge, guards must be installed, unless a fall arrest/restraint anchorage system is installed and in compliance with ANSI/ASSE Z 359.1. Additionally, the Occupational Safety and Health Administration's (OSHA) requires the use of fall protection or personal fall arrest equipment should maintenance staff need to access the roof area or equipment within 15-feet of the roof edge. Gale recommends considering placing ballasted perimeter guards along the edges of the roof where mechanical equipment (fans/vents) and roof access is within 15-feet of the roof edge. As such, Gale has provided a budgetary cost to install sections of the temporary ballasted guard rails, which will rest directly on the roof surface, around portions of designated roof

areas. Note, the use of the “temporary” rail system is to reduce potential significant costs of securing “permanent” rails into the structural framing of the existing roof deck, as well as to reduce the number of penetrations through the membrane, which could result in future leaks within the building. These rails are typically not intended to wrap around the entire building; however, portions of the rails will be visible from certain vantage points at the ground level.

Exterior Walls

Chapter 14 (IBC): Exterior Walls – The brick masonry renovations which have been recommended within this report will be performed in accordance with the MSBC installation requirements, as well as the current design industry standards as established by the Brick Industry Association (BIA).

Drainage

Chapter 15 (IBC): Roof Assemblies and Rooftop Structures – This section relates to drainage requirements for roof assemblies.

Section 1503.4 Roof Drainage – A review of the existing drainage capacity was performed for the facility. The majority of the existing roof drains throughout the low-sloped roof areas appear to be 4-inch diameter lead and oakum configurations. Gale reviewed the capacity of the existing roof drainage system for compliance with the current Massachusetts State Plumbing Code, 248 CMR 10.00 Uniform State Plumbing Code. Based on our calculations, the vertical leader capacity for the individual roof areas using these dimensions appears to be sufficient based on the Massachusetts State Plumbing Code requirements. Refer to the calculation sheets in the appendix of this report for additional information. As the roof drain leaders currently have a 3-inch copper sleeve inserts which do not meet code requirements, it is anticipated that these units were installed when the current drain bowls were installed in order to bridge the gap between the bowl and the leader pipe. It is the intent of this design to extend the existing vertical leaders and install new drain bowls with extension collars to provide the required insulation height at the sump locations. Therefore, access to the underside (below the roof deck) of each roof drain is anticipated to be required. The drain bowls should however be cleaned prior to completion of the roof construction.

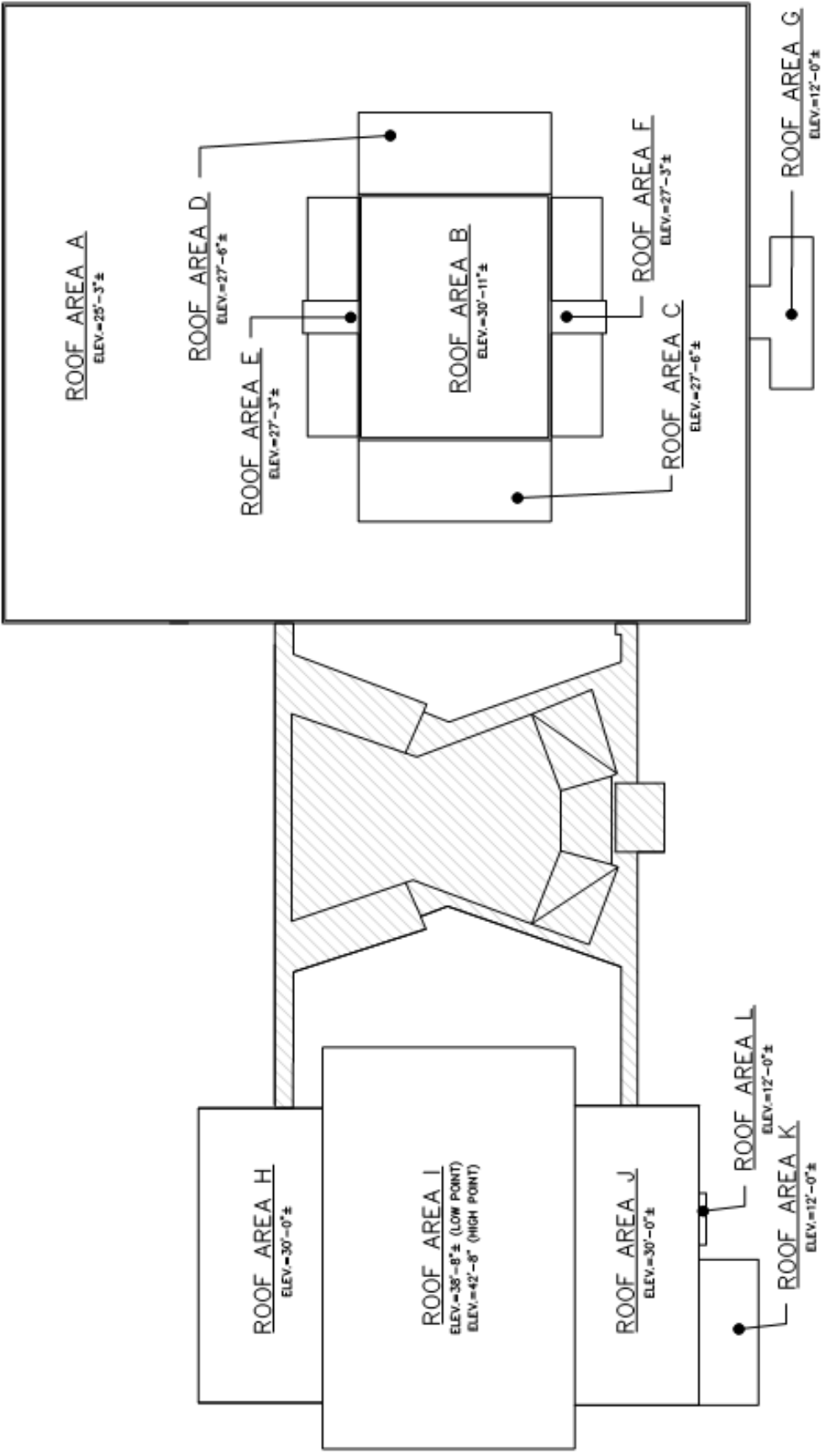
Appendix C

Engineering Calculations

Gale Associates, Inc.					
Drainage Calculation					
Prepared For: Amherst-Pelham Reginal School District					
Job Name: 3 Schools Roof/Boiler Project - Middle School					
833180					
Date: November 16, 2017			Prepared/Checked: CTM/SCM		
Capacity of Standard Vertical Leaders (In Square Feet of Roof Area) @ 4" per Hour					
Leader Dia. (Inches)	Roof Area (Sq. Ft.)		Leader Dia. (Inches)	Roof Area (Sq. Ft.)	
2	720		5	8650	
3	2200		6	13500	
4	4600		8	29000	
Capacity of Standard Horizontal Leaders (In Square Feet of Roof Area) @ 4" per Hour					
	1/8 inch slope	1/4 inch slope	1/2 inch slope		
Leader Dia. (Inches)	Roof Area (Sq. Ft.)	Roof Area (Sq. Ft.)	Roof Area (Sq. Ft.)		
3	0	1160	1644		
4	1880	2650	3760		
5	3340	4720	6680		
6	5350	7550	10700		
8	11500	16300	23000		
10	20700	29200	41400		
12	33300	47000	66600		
15	59500	84000	119000		
Capacities based on 4 inches per hour rain storm as required by the 248 CMR 10.00: Uniform State Plumbing Code					
Standard Roof Drainage Analysis (Capacities in Square Feet)					
Drainage Area Label	Roof Area (In Sq. Ft.)	Vertical Leader Capacity	Horizontal Leader Capacity	Actual Drainage Capacity	Pass/Fail
Area A	65,400	92,000	n/a	92,000	Pass
Area B	7,860	18,400	n/a	18,400	Pass
Area C	2,600	No Drain	n/a	Drains to Roof A	Pass
Area D	2,600	No Drain	n/a	Drains to Roof A	Pass
Area E	300	No Drain	n/a	Drains to Roof A	Pass
Area F	300	No Drain	n/a	Drains to Roof A	Pass
Area G	1,300	9,200	n/a	9,200	Pass
Area H	6,200	9,200	n/a	9,200	Pass
Area I	16,750	17,300	n/a	17,300	Pass
Area J	6,200	9,200	n/a	9,200	Pass
Area K	1,450	4,400	n/a	4,400	Pass
Area L	130	2,200	n/a	2,200	Pass
Total	105,290				
Description of Drainage Areas					
Drainage for Roof Area "A" is provided by (20) - 4" drain bowls.					
Drainage for Roof Area "B" is provided by (2) - 4" drain bowls.					

Roof Area "C" drains directly onto Roof Area "A"
Roof Area "D" drains directly onto Roof Area "A"
Roof Area "E" drains directly onto Roof Area "A"
Roof Area "F" drains directly onto Roof Area "A"
Drainage for Roof Area "G" is provided by (2) - 4" drain bowls (assumed).
Drainage for Roof Area "H" is provided by (2) - 4" drain bowls.
Drainage for Roof Area "I" is provided by (2) - 5" drain bowls.
Drainage for Roof Area "J" is provided by (2) - 4" drain bowls.
Drainage for Roof Area "K" is provided by (2) - 4" drain bowls.
Drainage for Roof Area "L" is provided by (1) - 4" scupper and downspout

Mass Building Code Section 1611.1 - Each portion of the roof shall be designed to sustain the load of rainwater that will accumulate on it if the primary drainage system for that portion is blocked plus the uniform load caused by water that rises above the inlet of the secondary drainage system at its design flow. This existing roof system currently incorporates overflow scuppers. Additional scuppers will be provided should the new insulation system obstruct this secondary drainage.



Gale Associates, Inc.					
Drainage Calculation					
Prepared For: Amherst-Pelham Reginal School District					
Job Name: 3 Schools Roof/Boiler Project - Middle School (Classroom Wing)					
Gale Job Number: 833180					
Date: November 30, 2017				Prepared/Checked: CTM/	
Capacity of Standard Horizontal Leaders (In Square Feet of Roof Area) @ 4" per Hour					
Leader Dia. (Inches)	1/8 Inch Slope Roof Area (Sq. Ft.)		1/4 Inch Slope Roof Area (Sq. Ft.)		1/2 Inch Slope Roof Area (Sq. Ft.)
3	0		1160		1644
4	1880		2650		3760
5	3340		4720		6680
6	5350		7550		10700
8	11500		16300		23000
10	20700		29200		41400
12	33300		47000		66600
15	59500		84000		119000
Capacities based on 4 inches per hour rain storm as required by the 248 CMR 10.00: Uniform State Plumbing Code					
Pipe Number (For Reference) (Bold Indicates Pipe Exits Building)	Approximate Drainage Area (Square Feet)	Pipe Diameter (inches).	Pass at 1/8"/ft. Slope?	Pass at 1/4"/ft. Slope?	Pass at 1/2"/ft. Slope?
1	2775	5	Y	Y	Y
2	2775	4	N	N	Y
3	5550	6	N	Y	Y
4	3253	5	Y	Y	Y
5	6506	8	Y	Y	Y
6	9760	8	Y	Y	Y
7	3564	4	N	N	Y
8	3564	5	N	Y	Y
9	7128	6	N	Y	Y
10	1875	4	Y	Y	Y
11	1875	4	Y	Y	Y
12	2621	6	Y	Y	Y
13	3125	6	Y	Y	Y
14	5746	6	N	Y	Y
15	3100	6	Y	Y	Y
16	3750	5	N	Y	Y
17	8846	8	Y	Y	Y
18	12596	8	N	Y	Y
19	2850	5	Y	Y	Y
20	15446	10	N	Y	Y
21	3100	6	Y	Y	Y
22	18546	10	N	Y	Y
23	1875	4	Y	Y	Y
24	1875	4	Y	Y	Y
25	3750	5	N	Y	Y
26	2650	5	Y	Y	Y
27	6400	5	N	N	Y
28	3250	5	Y	Y	Y
29	9650	6	N	N	Y
30	3250	6	Y	Y	Y
30A	2850	5	Y	Y	Y
31	15750	8	N	Y	Y
32	3320	5	Y	Y	Y
33	3250	5	Y	Y	Y
34	3250	5	Y	Y	Y
35	3250	5	Y	Y	Y
36	6506	6	N	Y	Y
37	9760	8	Y	Y	Y
38	13080	8	N	Y	Y

Gale Associates, Inc.					
Drainage Calculation					
Prepared For: Amherst-Pelham Reginal School District					
Job Name: 3 Schools Roof/Boiler Project - Middle School (Gym/Pool Wing)					
Gale Job Number: 833180					
Date: November 30, 2017			Prepared/Checked: CTM/		
Capacity of Standard Horizontal Leaders (In Square Feet of Roof Area) @ 4" per Hour					
Leader Dia. (Inches)	1/8 Inch Slope Roof Area (Sq. Ft.)		1/4 Inch Slope Roof Area (Sq. Ft.)		1/2 Inch Slope Roof Area (Sq. Ft.)
3	0		1160		1644
4	1880		2650		3760
5	3340		4720		6680
6	5350		7550		10700
8	11500		16300		23000
10	20700		29200		41400
12	33300		47000		66600
15	59500		84000		119000
Capacities based on 4 inches per hour rain storm as required by the 248 CMR 10.00: Uniform State Plumbing Code					
Pipe Number (For Reference) (Bold Indicates Pipe Exits Building)	Approximate Drainage Area (Square Feet)	Pipe Diameter (inches).	Pass at 1/8"/ft. Slope?	Pass at 1/4"/ft. Slope?	Pass at 1/2"/ft. Slope?
1	3050	5	Y	Y	Y
2	3050	5	Y	Y	Y
3	8375	8	Y	Y	Y
4	8375	8	Y	Y	Y
5	3050	5	Y	Y	Y
6	3050	5	Y	Y	Y
7	3050	5	Y	Y	Y
8	3050	5	Y	Y	Y
9	3050	8	Y	Y	Y
10	6100	8	Y	Y	Y
11	14475	10	Y	Y	Y
12	8375	8	Y	Y	Y
13	22850	10	N	Y	Y
14	25900	12	Y	Y	Y
15	28950	12	Y	Y	Y

**GALE ASSOCIATES INC**163 LIBBEY PARKWAY
WEYMOUTH, MA 02189
(781) 335-6465JOB 833180 - AMHERST MIDDLE SCHOOL
SHEET NO. 1 OF 2
CALCULATED BY CTM DATE 11/13/17
CHECKED BY _____ DATE _____
SCALE 3" = 1'-0"WIND LOADS - AMHERST REGIONAL MIDDLE SCHOOL(ROOF AREA I)CODES: IBC 2015
IRC 2015
ASCE 7-10
MSRC 9TH EDITION

CITY: AMHERST, MA

HEIGHT: 45'

LENGTH: 165'-4" (PER ROOF AREA I)

WIDTH: 101'-4" (PER ROOF AREA I)

LOW RISE BUILDING (L+L)(ASCE 7-10)

STEP 1: RISK CATEGORY III

STEP 2: $V_{WT} = 125$ mphSTEP 3: $K_d = 0.85$

EXPOSURE CATEGORY C

 $K_{zt} = 1$

ENCLOSED

 $GCP_i = \pm 0.18$ STEP 4: $K_2 = 2.01 \left[\frac{Z}{Z_0} \right]^{2.5}$

$$K_2 = 2.01 \left[\frac{45}{900} \right]^{2.5}$$

$$K_2 = 1.07$$

STEP 5: $q_h = 0.00256 K_2 K_{zt} K_d V^2$

$$q_h = 0.00256 (1.07)(1)(0.85)(125)^2$$

$$q_h = 36 \text{ psf}$$

(TABLE 1.5-1)(MSRC 9TH)(TABLE 26.6-1)(SECT. 26.7)(FIG 26.8-1)(SECT. 26.10)(TABLE 26.11-1)(TABLE 30.3-1)

**GALE ASSOCIATES INC**

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(781) 335-6465

JOB _____

SHEET NO. 2 OF 2

CALCULATED BY _____ DATE _____

CHECKED BY _____ DATE _____

SCALE _____

STEP 6:

(FIG. 30.4-2A)

$$GCP_i = 0.18 \text{ (ENCLOSED BUILDING)}$$

$$\text{ZONE 1 (FIELD)} : GCP = -1.0$$

$$P = qh [GCP - GCP_i] = 36 [-1.0 - (0.18)] = P = -42.48 \text{ psf}$$

$$\text{ZONE 2 (PERIMETER)} : GCP = -1.8$$

$$P = qh [GCP - GCP_i] = 36 [-1.8 - (0.18)] = P = -71.28 \text{ psf}$$

$$\text{ZONE 3 (CORNER)} : GCP = -2.8$$

$$P = qh [GCP - GCP_i] = 36 [-2.8 - (0.18)] = P = -107.28 \text{ psf}$$

Untitled Map

Write a description for your map.

Legend

Line Measure

Google earth

2000 ft

N





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Project
APRSD Middle School Wind Uplift @ Pool Bldg

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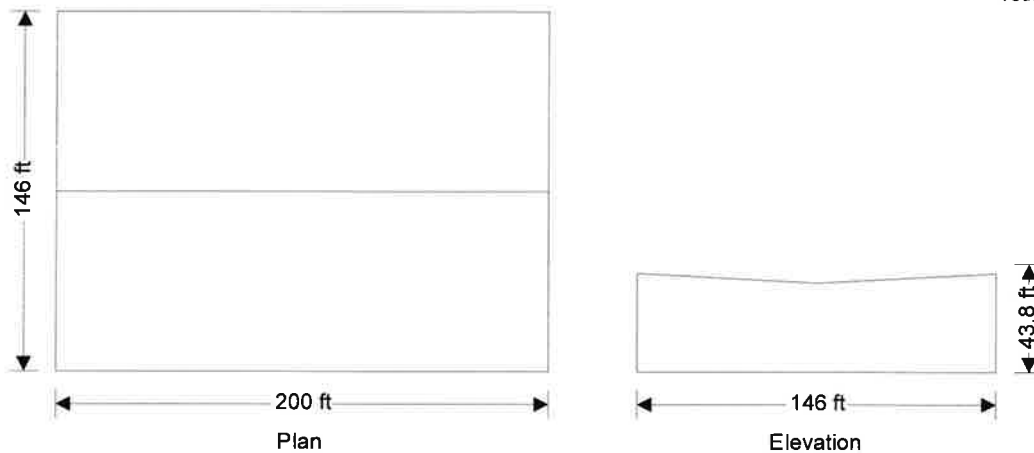
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WIND LOADING (ASCE7-10)

In accordance with ASCE7-10 incorporating Errata No. 1 and Errata No. 2

Using the components and cladding design method

Tedds calculation version 2.0.15



Building data

Type of roof	Gable
Length of building	b = 200.00 ft
Width of building	d = 146.00 ft
Height to eaves	H = 40.00 ft
Pitch of roof	$\alpha_0 = 3.0$ deg
Mean height	h = 40.00 ft

General wind load requirements

Basic wind speed	V = 125.0 mph
Risk category	III
Velocity pressure exponent coeff (Table 26.6-1)	$K_d = 0.85$
Exposure category (cl.26.7.3)	C
Enclosure classification (cl.26.10)	Enclosed buildings
Internal pressure coef +ve (Table 26.11-1)	$GC_{pi,p} = 0.18$
Internal pressure coef -ve (Table 26.11-1)	$GC_{pi,n} = -0.18$
Gust effect factor	$G_f = 0.85$

Topography

Topography factor not significant	$K_{zt} = 1.0$
-----------------------------------	----------------

Velocity pressure

Velocity pressure coefficient (T.30.3-1)	$K_z = 1.04$
Velocity pressure	$q_h = 0.00256 \times K_z \times K_{zt} \times K_d \times V^2 \times 1 \text{ psf/mph}^2 = 35.4 \text{ psf}$

Peak velocity pressure for internal pressure

Peak velocity pressure – internal (as roof press.)	$q_i = 35.36 \text{ psf}$
--	---------------------------

Equations used in tables

Net pressure	$p = q_h \times [GC_p - GC_{pi}]$
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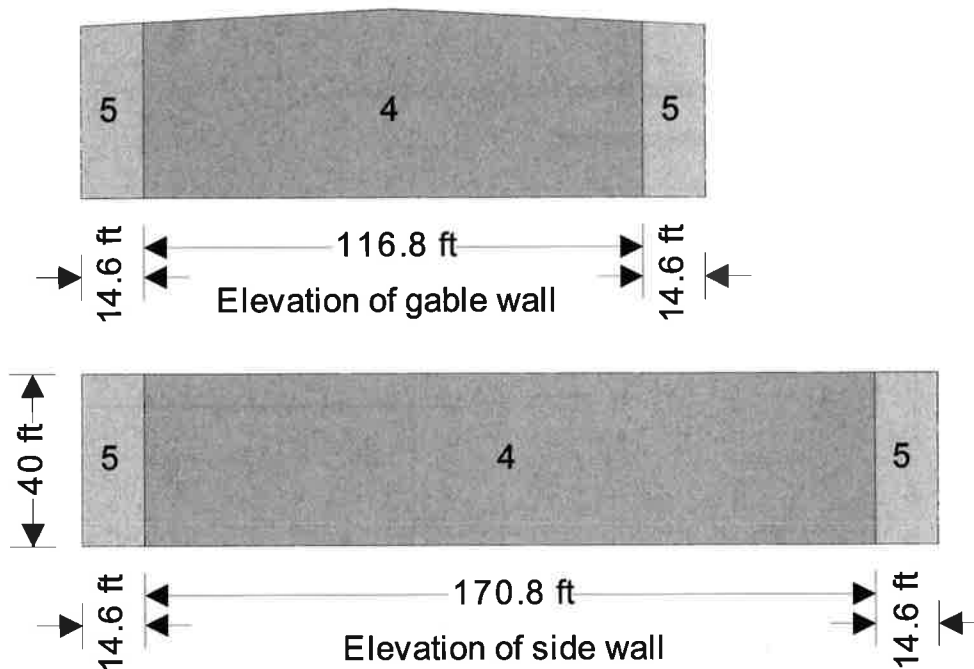
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Components and cladding pressures - Roof (Figure 30.4-2A)

Component	Zone	Length (ft)	Width (ft)	Eff. area (ft ²)	+GC _p	-GC _p	Pres (+ve) (psf)	Pres (-ve) (psf)
<10sf	1	-	-	10.0	0.30	-1.00	17.0	-41.7
25sf	1	-	-	25.0	0.26	-0.96	15.6 #	-40.3
50sf	1	-	-	50.0	0.23	-0.93	14.5 #	-39.3
>100sf	1	-	-	100.0	0.20	-0.90	13.4 #	-38.2
<10sf	2	-	-	10.0	0.30	-1.80	17.0	-70.0
25sf	2	-	-	25.0	0.26	-1.52	15.6 #	-60.2
50sf	2	-	-	50.0	0.23	-1.31	14.5 #	-52.7
>100sf	2	-	-	100.0	0.20	-1.10	13.4 #	-45.3
<10sf	3	-	-	10.0	0.30	-2.80	17.0	-105.4
25sf	3	-	-	25.0	0.26	-2.12	15.6 #	-81.5
50sf	3	-	-	50.0	0.23	-1.61	14.5 #	-63.4
>100sf	3	-	-	100.0	0.20	-1.10	13.4 #	-45.3

The final net design wind pressure, including all permitted reductions, used in the design shall not be less than 16psf acting in either direction



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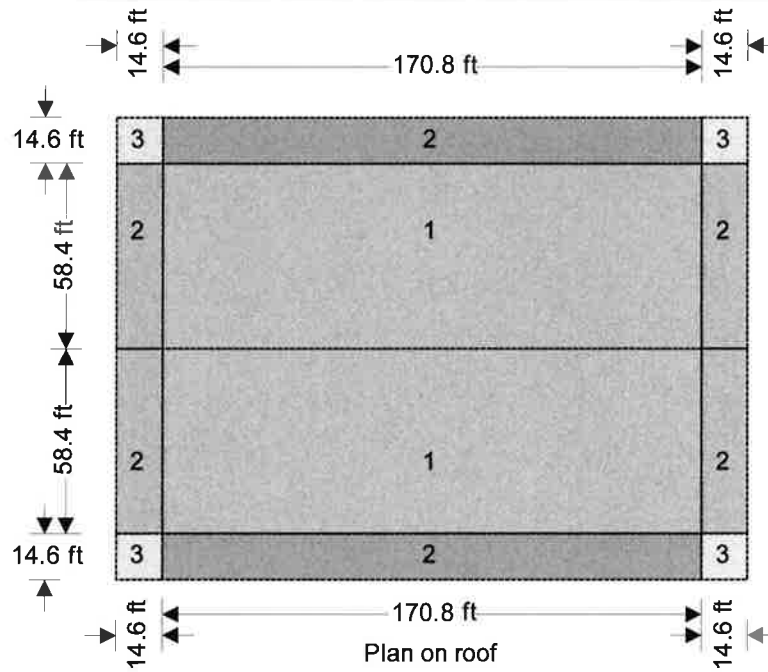
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APRSD Snow Load Analysis @ Pool/Gym Bldg.

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Job Ref.
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1

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SNOW LOADING (ASCE7-10)

Tedds calculation version 1.0.05

Building details

Roof type Flat
Width of roof $b = 164.00$ ft

Ground snow load

Ground snow load $p_g = 40.00$ lb/ft²
Density of snow $\gamma = \min(0.13 \times p_g / 1 \text{ ft} + 14 \text{ lb/ft}^3, 30 \text{ lb/ft}^3) = 19.20$ lb/ft³
Terrain type C
Exposure condition (Table 7-2) Fully exposed
Exposure factor (Table 7-2) $C_e = 0.90$
Thermal condition (Table 7-3) Structures kept just above freezing
Thermal factor (Table 7-3) $C_t = 1.10$
Importance category (Table 1-1) III
Importance factor (Table 7-4) $I_s = 1.10$
Min snow load for low slope roofs (Sect 7.3.4) $p_{f_min} = I_s \times 20 \text{ lb/ft}^2 = 22.00$ lb/ft²
Flat roof snow load (Sect 7.3) $p_r = 0.7 \times C_e \times C_t \times I_s \times p_g = 30.49$ lb/ft²

Balanced load

30.5 psf



Roof elevation

Drift calculations

Balanced snow load height $h_b = p_r / \gamma = 1.59$ ft
Length of upper roof $l_u = 100.00$ ft
Length of lower roof $l_l = 50.00$ ft
Height diff between upper and lower roofs $h_{diff} = 13.00$ ft
Height from balance load to top of upper roof $h_c = h_{diff} - h_b = 11.41$ ft
Drift height leeward drift $h_{d_l} = 0.43 \times (\max(20 \text{ ft}, l_u) \times 1 \text{ ft}^2)^{1/3} \times (p_g / 1 \text{ lb/ft}^2 + 10)^{1/4} - 1.5 \text{ ft} = 3.81$ ft
Drift height windward drift $h_{d_w} = 0.75 \times (0.43 \times (\max(20 \text{ ft}, l_l) \times 1 \text{ ft}^2)^{1/3} \times (p_g / 1 \text{ lb/ft}^2 + 10)^{1/4} - 1.5 \text{ ft}) = 2.03$ ft
Maximum lw/ww drift height $h_{d_max} = \max(h_{d_w}, h_{d_l}) = 3.81$ ft
Drift height $h_d = \min(h_{d_max}, h_c) = 3.81$ ft
Drift width $W_d = \min(4 \times h_d, 8 \times h_c) = 15.23$ ft
Drift surcharge load $p_d = h_d \times \gamma = 73.10$ lb/ft²



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APRSD Snow Load Analysis @ Pool/Gym Bldg.

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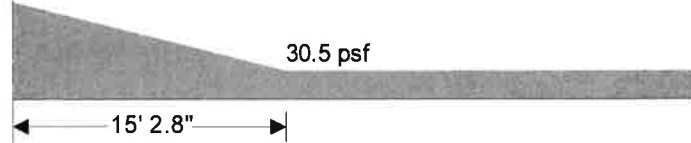
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103.6 psf



Elevation on snow drift



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Project APRSD Snow Load Analysis @ Classroom Bldg.				Job Ref. 833180	
Section				Sheet no./rev. 1	
Calc. by SMF	Date 11/29/2017	Chk'd by	Date	App'd by	Date

SNOW LOADING (ASCE7-10)

Tedds calculation version 1.0.05

Building details

Roof type Flat
Width of roof $b = 254.00$ ft

Ground snow load

Ground snow load $p_g = 40.00$ lb/ft²
Density of snow $\gamma = \min(0.13 \times p_g / 1 \text{ ft} + 14 \text{ lb/ft}^3, 30 \text{ lb/ft}^3) = 19.20$ lb/ft³
Terrain type C
Exposure condition (Table 7-2) Fully exposed
Exposure factor (Table 7-2) $C_e = 0.90$
Thermal condition (Table 7-3) Structures kept just above freezing
Thermal factor (Table 7-3) $C_t = 1.10$
Importance category (Table 1-1) III
Importance factor (Table 7-4) $I_s = 1.10$
Min snow load for low slope roofs (Sect 7.3.4) $p_{r_min} = I_s \times 20 \text{ lb/ft}^2 = 22.00$ lb/ft²
Flat roof snow load (Sect 7.3) $p_r = 0.7 \times C_e \times C_t \times I_s \times p_g = 30.49$ lb/ft²

Roof projection drifts

Max height of obstruction $h_{obs} = \max(h_{1_obs}, h_{2_obs}) = 3.00$ ft
Width of obstruction $b_{obs} = 96.00$ ft
Distance from LHS eaves $b_{1_obs} = 79.00$ ft
Distance from RHS eaves $b_{2_obs} = 79.00$ ft
Balanced snow load height $h_b = p_r / \gamma = 1.59$ ft
Height from balance load to top of projection $h_{c_obs} = h_{obs} - h_b = 1.41$ ft
Length of lower roof $l_{l_obs} = \max(b_{1_obs}, b_{2_obs}) = 79.00$ ft
Drift height windward drift $h_{d_l_obs} = 0.75 \times (0.43 \times (\max(20 \text{ ft}, l_{l_obs}) \times 1 \text{ ft}^2)^{1/3} \times (p_g / 1 \text{ lb/ft}^2 + 10)^{1/4} - 1.5 \text{ ft}) = 2.55$ ft
Drift height $h_{d_obs} = \min(h_{d_l_obs}, h_{obs} - h_b) = 1.41$ ft
Drift width $W_{d_obs} = \min(4 \times h_{d_l_obs}, 8 \times (h_{obs} - h_b)) = 10.22$ ft
Drift surcharge load $p_{d_obs} = h_{d_obs} \times \gamma = 27.11$ lb/ft²



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Project
APRSD Snow Load Analysis @ Classroom Bldg.

Job Ref.
833180

Section

Sheet no./rev.
2

Calc. by
SMF

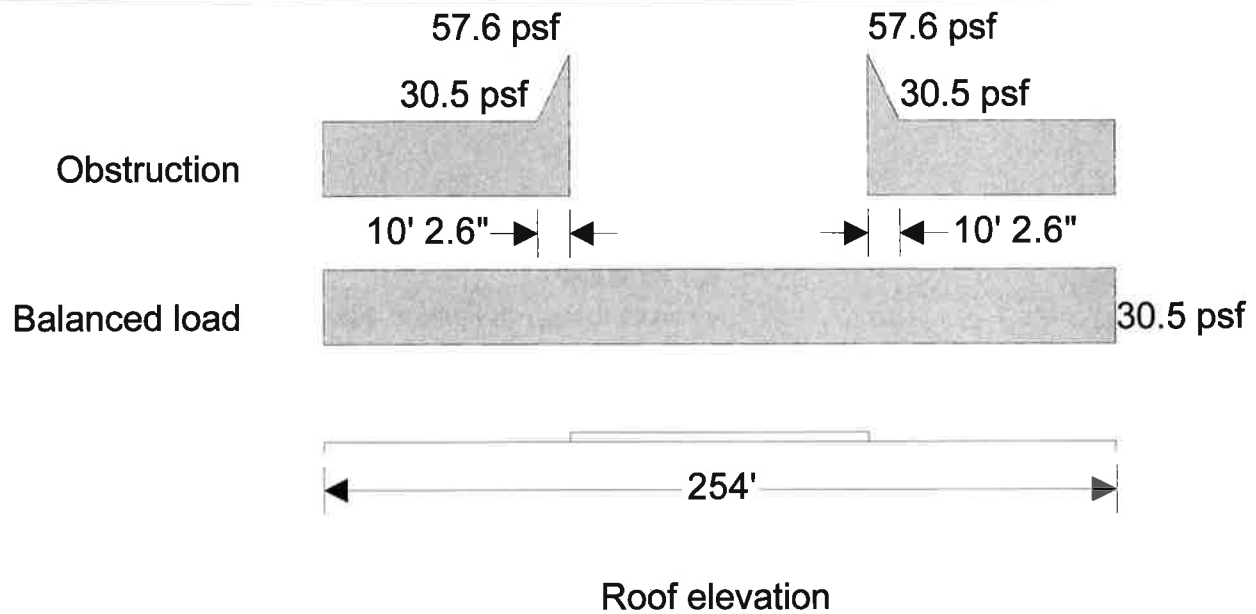
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11/29/2017

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Date





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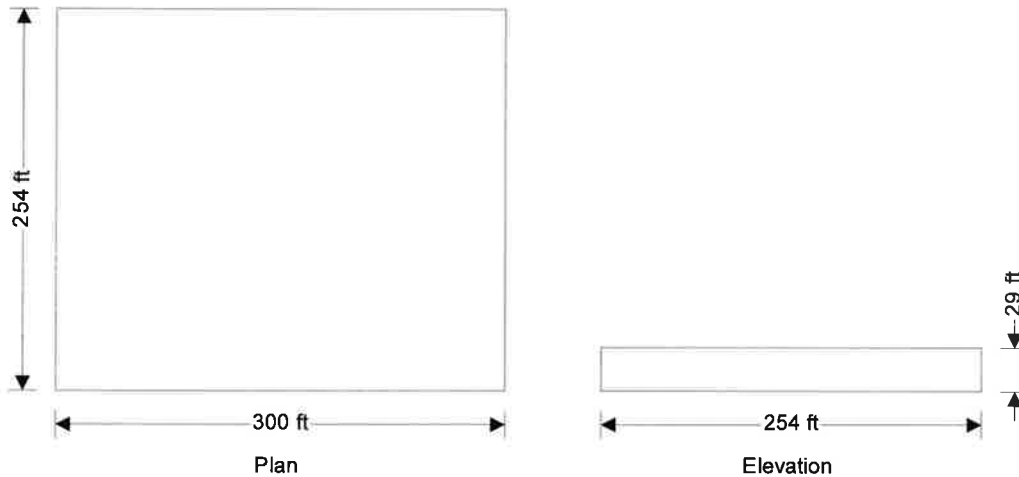
Project APRSD Middle School Wind Uplift @ Classroom Bldg				Job Ref. 833180	
Section				Sheet no./rev. 1	
Calc. by SMF	Date 11/29/2017	Chk'd by	Date	App'd by	Date

WIND LOADING (ASCE7-10)

In accordance with ASCE7-10 incorporating Errata No. 1 and Errata No. 2

Using the components and cladding design method

Tedds calculation version 2.0.15



Building data

Type of roof	Flat
Length of building	b = 300.00 ft
Width of building	d = 254.00 ft
Height to eaves	H = 29.00 ft
Mean height	h = 29.00 ft

General wind load requirements

Basic wind speed	V = 125.0 mph
Risk category	III
Velocity pressure exponent coeff (Table 26.6-1)	K _d = 0.85
Exposure category (cl.26.7.3)	C
Enclosure classification (cl.26.10)	Enclosed buildings
Internal pressure coef +ve (Table 26.11-1)	GC _{pi_p} = 0.18
Internal pressure coef -ve (Table 26.11-1)	GC _{pi_n} = -0.18
Gust effect factor	G _f = 0.85

Topography

Topography factor not significant	K _{zt} = 1.0
-----------------------------------	------------------------------

Velocity pressure

Velocity pressure coefficient (T.30.3-1)	K _z = 0.97
Velocity pressure	q _h = 0.00256 × K _z × K _{zt} × K _d × V ² × 1psf/mph ² = 33.0 psf

Peak velocity pressure for internal pressure

Peak velocity pressure – internal (as roof press.)	q _i = 33.05 psf
--	-----------------------------------

Equations used in tables

Net pressure	p = q _h × [GC _p - GC _{pi}]
--------------	--



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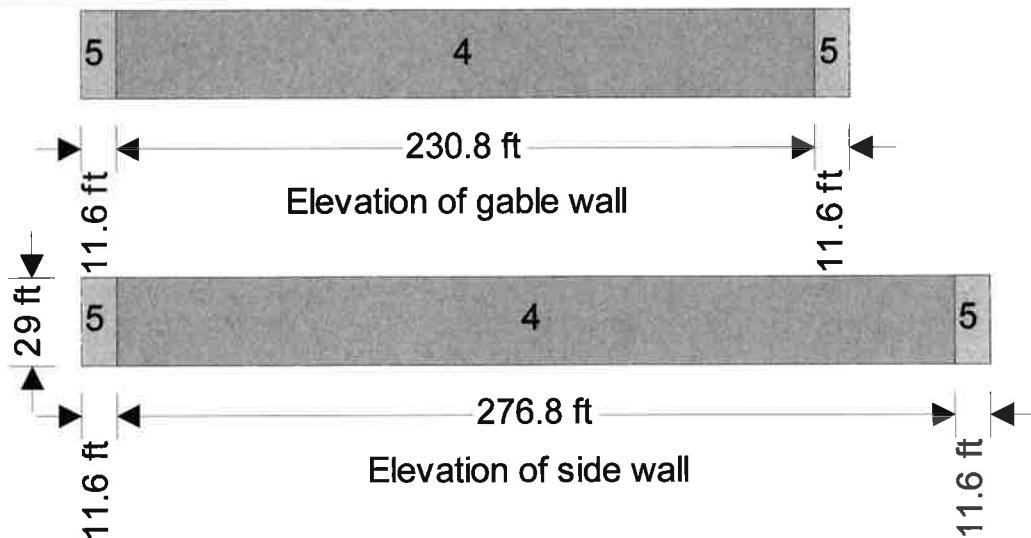
Date
11/29/2017

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Components and cladding pressures - Roof (Figure 30.4-2A)

Component	Zone	Length (ft)	Width (ft)	Eff. area (ft ²)	+GC _p	-GC _p	Pres (+ve) (psf)	Pres (-ve) (psf)
<10sf	1	-	-	10.0	0.30	-1.00	15.9 #	-39.0
25sf	1	-	-	25.0	0.26	-0.96	14.5 #	-37.7
50sf	1	-	-	50.0	0.23	-0.93	13.6 #	-36.7
>100sf	1	-	-	100.0	0.20	-0.90	12.6 #	-35.7
<10sf	2	-	-	10.0	0.30	-1.80	15.9 #	-65.4
25sf	2	-	-	25.0	0.26	-1.52	14.5 #	-56.2
50sf	2	-	-	50.0	0.23	-1.31	13.6 #	-49.3
>100sf	2	-	-	100.0	0.20	-1.10	12.6 #	-42.3
<10sf	3	-	-	10.0	0.30	-2.80	15.9 #	-98.5
25sf	3	-	-	25.0	0.26	-2.12	14.5 #	-76.1
50sf	3	-	-	50.0	0.23	-1.61	13.6 #	-59.2
>100sf	3	-	-	100.0	0.20	-1.10	12.6 #	-42.3

The final net design wind pressure, including all permitted reductions, used in the design shall not be less than 16psf acting in either direction



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APRSD Middle School Wind Uplift @ Classroom Bldg

Job Ref.

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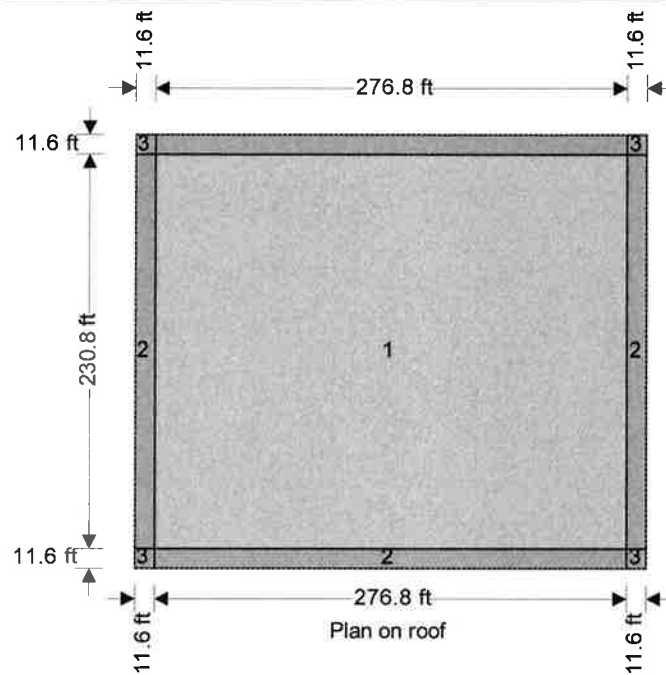
11/29/2017

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Date





Appendix D

Hazardous Material Testing



EMSL Analytical, Inc.

5 Constitution Way, Unit A Woburn, MA 01801

Tel/Fax: (781) 933-8411 / (781) 933-8412

<http://www.EMSL.com / bostonlab@emsl.com>

EMSL Order: 131705192

Customer ID: GALA25

Customer PO: 1510670506

Project ID:

Attention: Cole McCarthy
Gale Associates, Inc.
163 Libbey Industrial Parkway
Weymouth, MA 02189

Phone: (781) 335-6465

Fax: (781) 335-6467

Received Date: 11/15/2017 9:08 AM

Analysis Date: 11/16/2017

Collected Date: 11/09/2017

Project: 833180 Amherst Middle School

Test Report: Asbestos Analysis of Bulk Materials via EPA 600/R-93/116 Method using Polarized Light Microscopy

Sample	Description	Appearance	Non-Asbestos		Asbestos
			% Fibrous	% Non-Fibrous	% Type
1 131705192-0001	Cont. Sealant @ TWF of Roof Area 'A' to Area 'B'	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
2 131705192-0002	Roof Area 'A' TWF Sealant @ Rising Wall (TC1)	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
3 131705192-0003	Sealant Above Flashing of Roof Area 'F' to 'B'	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
4 131705192-0004	Roof Area 'A' Antenna Tension Wire Post Sealant	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
5 131705192-0005	Test Cut 13 Asphaltic Material on Deck	Brown/Black Fibrous Homogeneous	50% Cellulose	50% Non-fibrous (Other)	None Detected
6 131705192-0006	Test Cut 12 Sealant Above TWF @ Rising Wall	Black Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
7 131705192-0007	Test Cut 14 Sealant at Chimney TWF	Brown Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected
8 131705192-0008	Drain Leader Insulation	White/Yellow Fibrous Homogeneous	80% Glass	20% Non-fibrous (Other)	None Detected
9 131705192-0009	Roof Area K Rising Wall Sealant	Gray Non-Fibrous Homogeneous		100% Non-fibrous (Other)	None Detected

Analyst(s)

Elizabeth Stutts (9)

Steve Grise, Laboratory Manager
or Other Approved Signatory

EMSL maintains liability limited to cost of analysis. This report relates only to the samples reported and may not be reproduced, except in full, without written approval by EMSL. EMSL bears no responsibility for sample collection activities or analytical method limitations. Interpretation and use of test results are the responsibility of the client. This report must not be used by the client to claim product certification, approval, or endorsement by NVLAP, NIST or any agency of the federal government. Non-friable organically bound materials present a problem matrix and therefore EMSL recommends gravimetric reduction prior to analysis. Samples received in good condition unless otherwise noted. Estimated accuracy, precision and uncertainty data available upon request. Unless requested by the client, building materials manufactured with multiple layers (i.e. linoleum, wallboard, etc.) are reported as a single sample. Reporting limit is 1%

Samples analyzed by EMSL Analytical, Inc. Woburn, MA NVLAP Lab Code 101147-0, CT PH-0315, MA AA000188, RI AAL-107T3, VT AL998919, Maine Bulk Asbestos BA039

Initial report from: 11/17/2017 07:25:30



Appendix E

Reduced Drawings

ROOF REPLACEMENT AND ASSOCIATED WORK AT THE

AMHERST REGIONAL MIDDLE SCHOOL

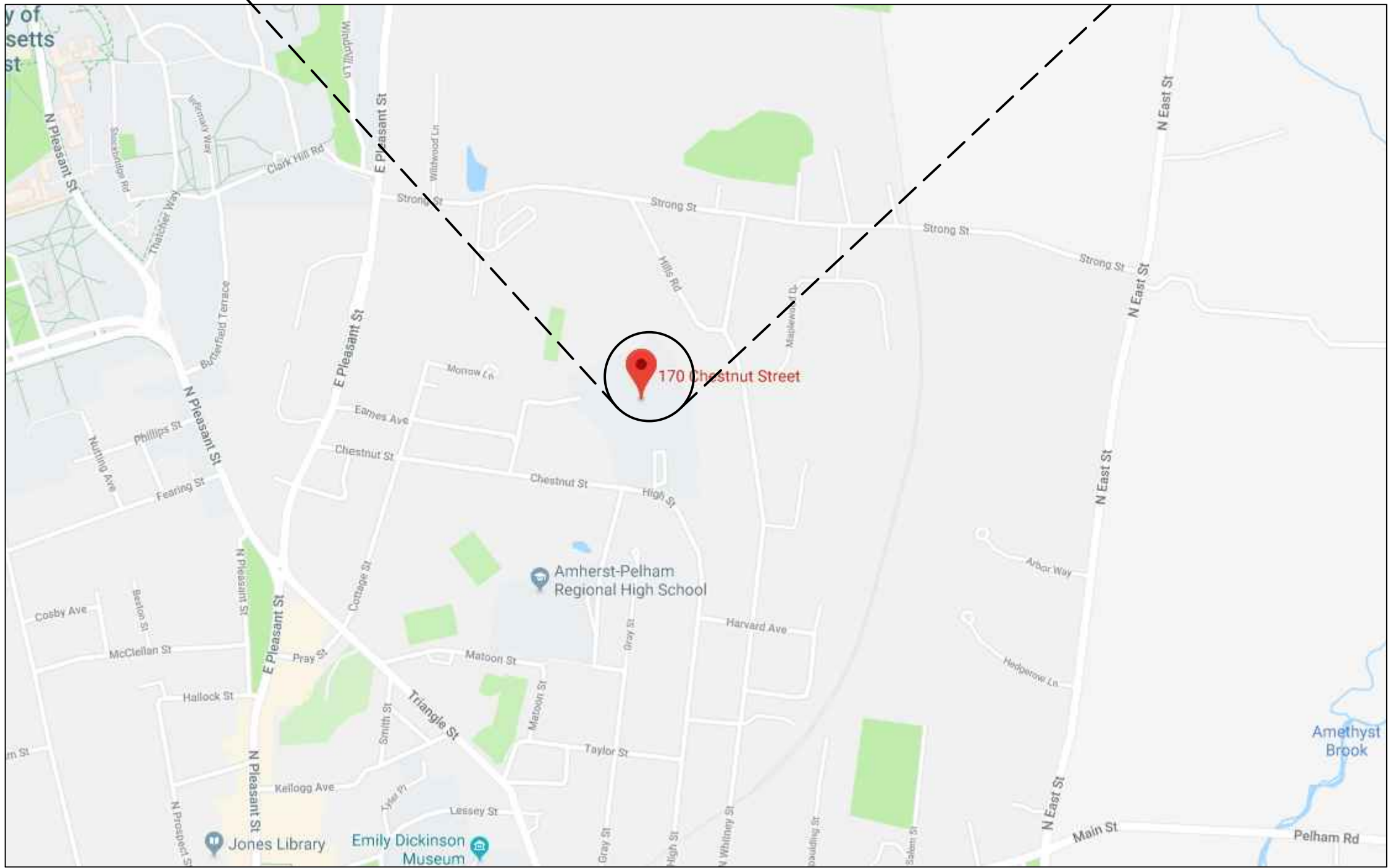
170 CHESTNUT ST
AMHERST, MA 01002

PREPARED FOR
AMHERST-PELHAM REGIONAL SCHOOL DISTRICT
170 CHESTNUT STREET
AMHERST, MA 01002

DRAWING NO	TITLE
G100	COVER SHEET
G101	EXISTING AND NEW ROOF CROSS SECTIONS AND GENERAL NOTES
G102	SITE PLAN AND NOTES
G103	CONCEPTUAL TAPERED INSULATION PLAN AND NOTES
A101	CLASSROOM WING ROOF AREA PLAN
A102	GYM/POOL ROOF AREA PLAN



SITE MAP
NOT TO SCALE



LOCATION MAP
NOT TO SCALE



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PROJECT	ROOF REPLACEMENT AND ASSOCIATED WORK AT THE AMHERST REGIONAL MIDDLE SCHOOL AMHERST, MA 01002
	OWNER AMHERST-PELHAM REGIONAL SCHOOL DISTRICT 170 CHESTNUT STREET AMHERST, MA 01002

NO.	DATE	DESCRIPTION	BY
PROJECT NO.	833180		
CADD FILE	833180 G100		
DESIGNED BY	CTM		
DRAWN BY	RRF		
CHECKED BY	CM		
DATE	12/08/2017		
DRAWING SCALE	NOT TO SCALE		

GRAPHIC SCALE

SHEET TITLE

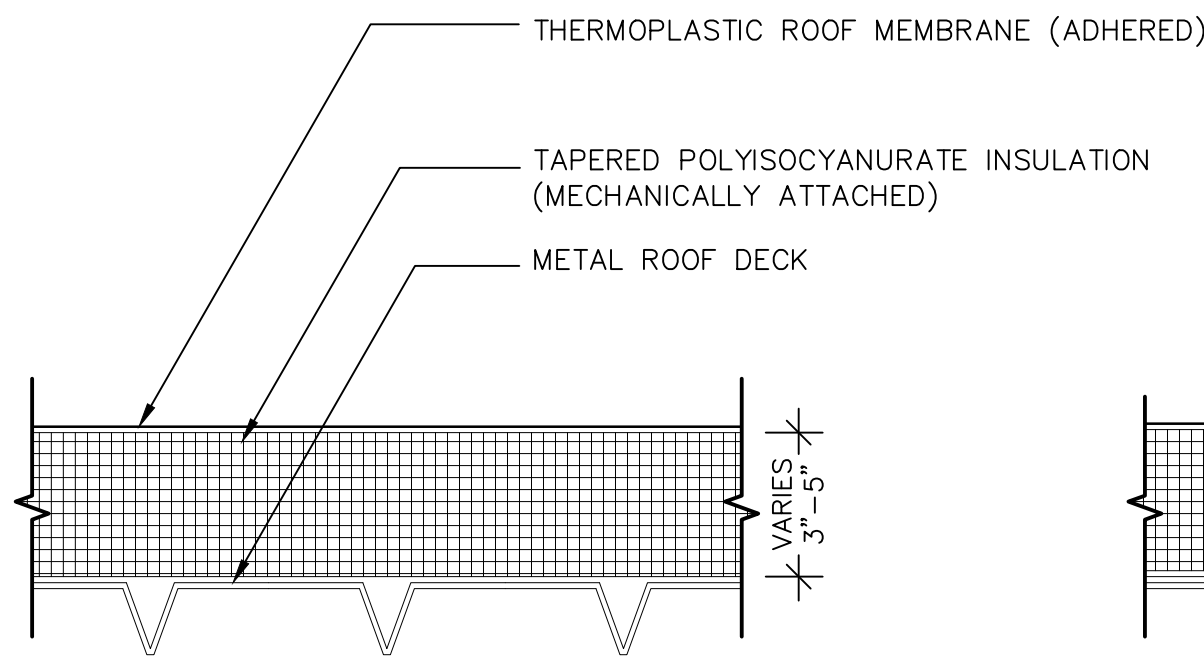
COVER SHEET

DRAWING NO.

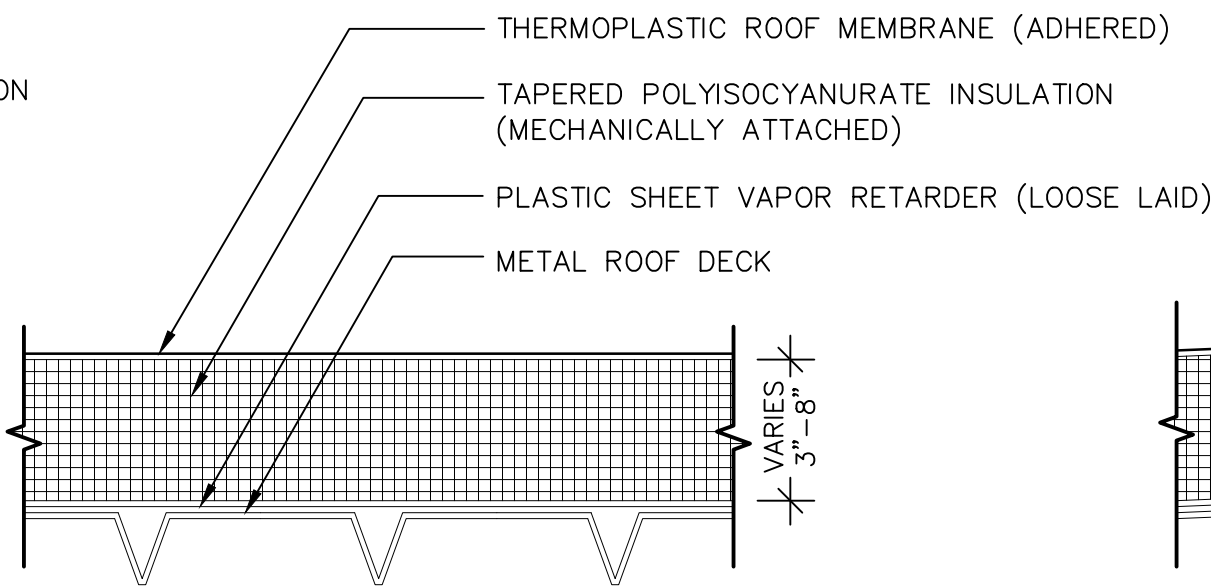
G100

EXISTING ROOF CROSS SECTIONS

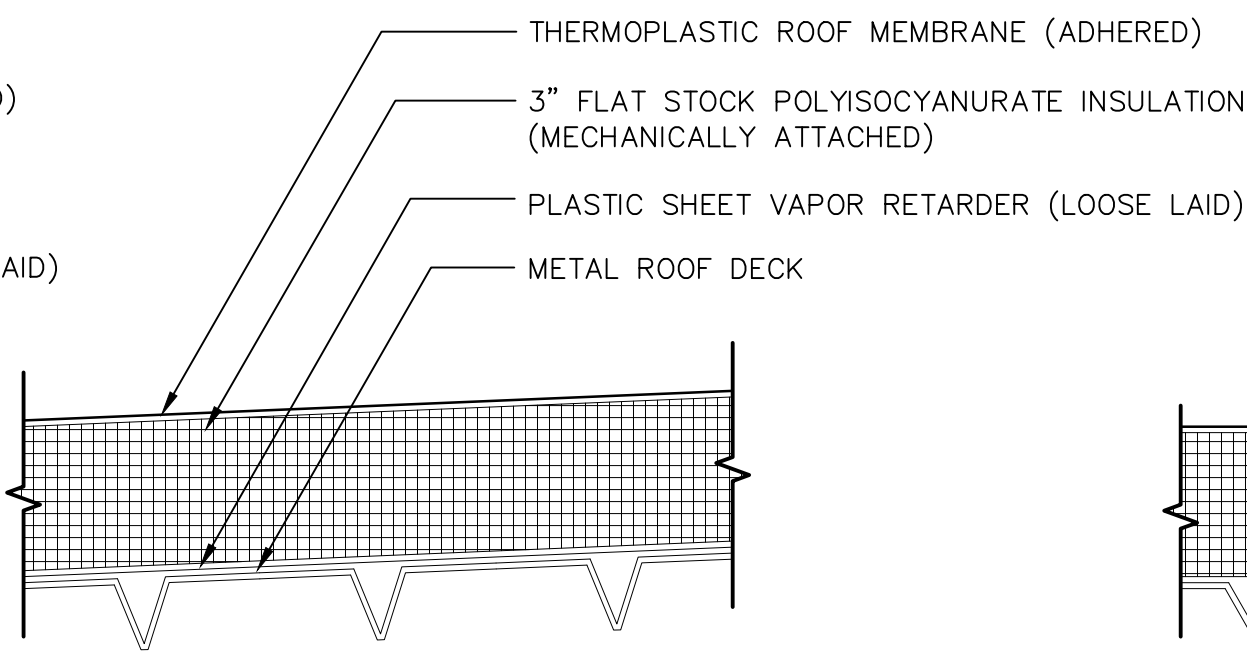
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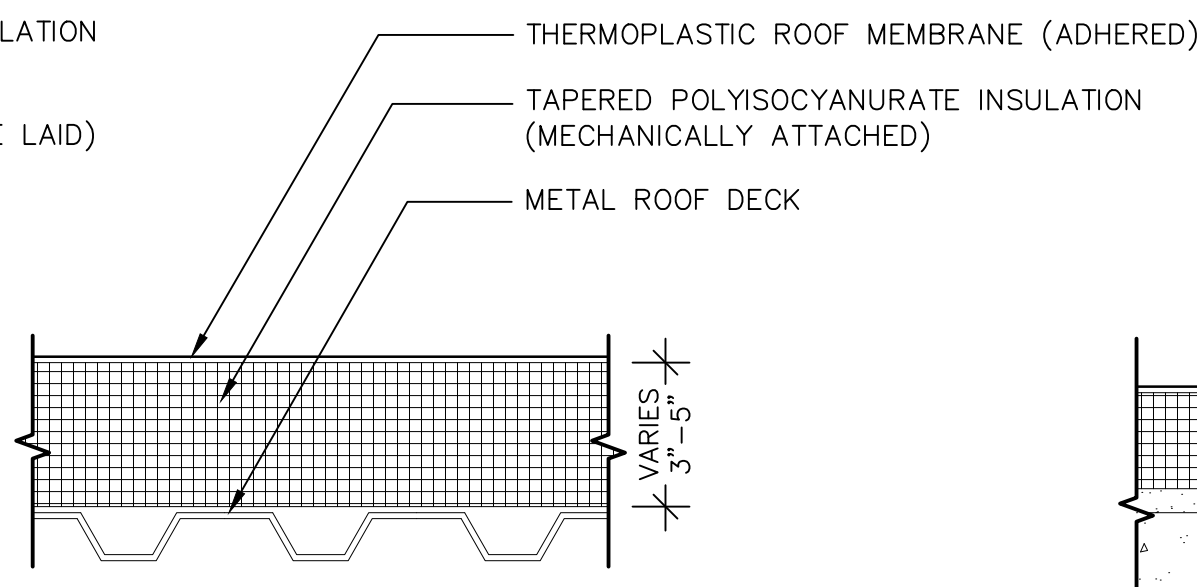
ROOF AREAS A, B, AND L



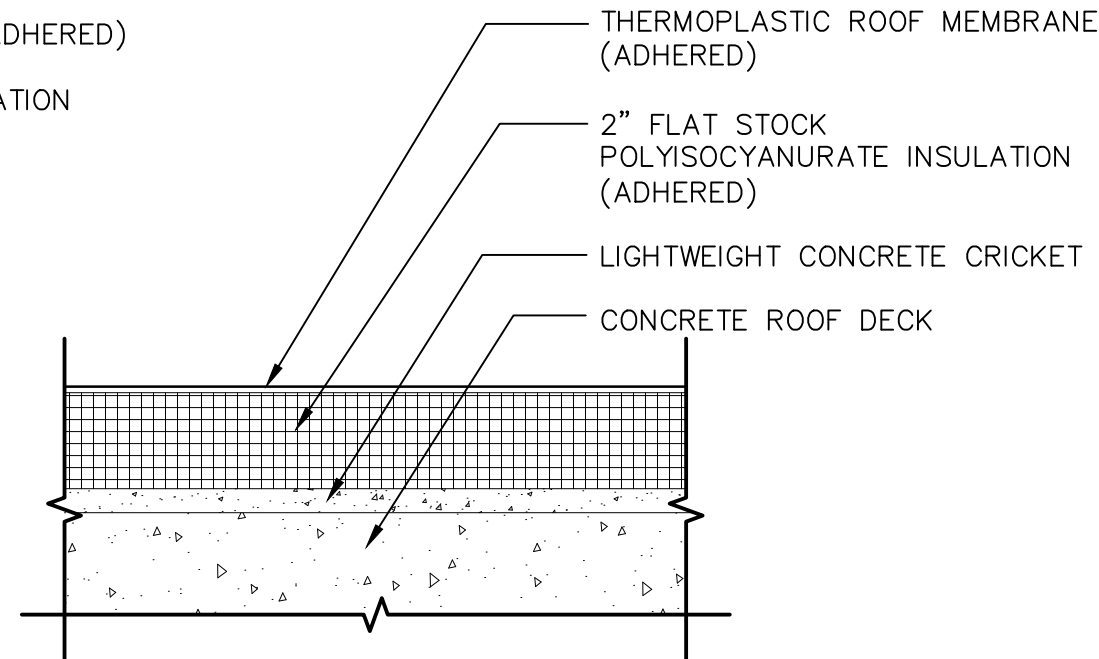
ROOF AREAS H, AND J



ROOF AREA I



ROOF AREAS C, D, E, F, AND K



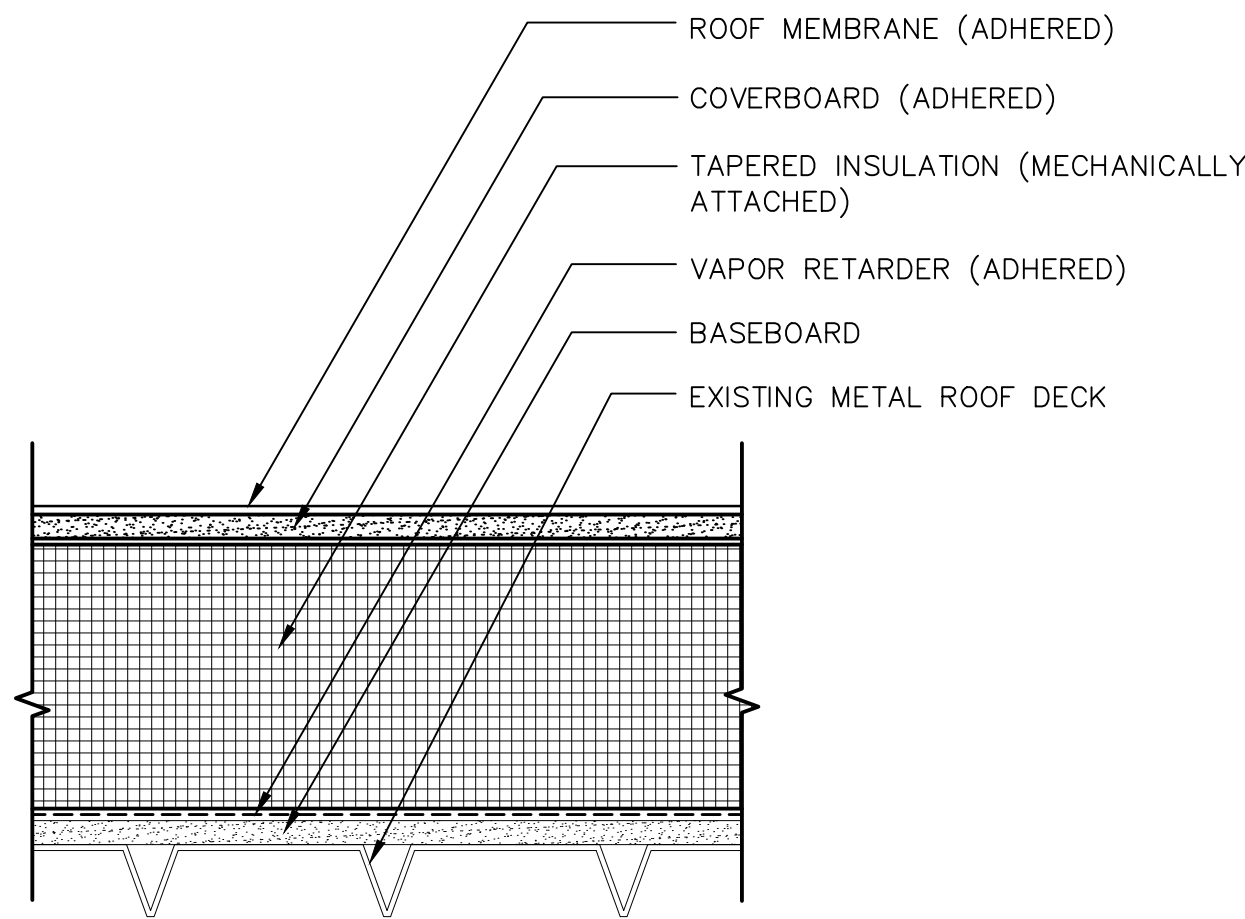
ROOF AREA G

PROPOSED ROOF CROSS SECTIONS

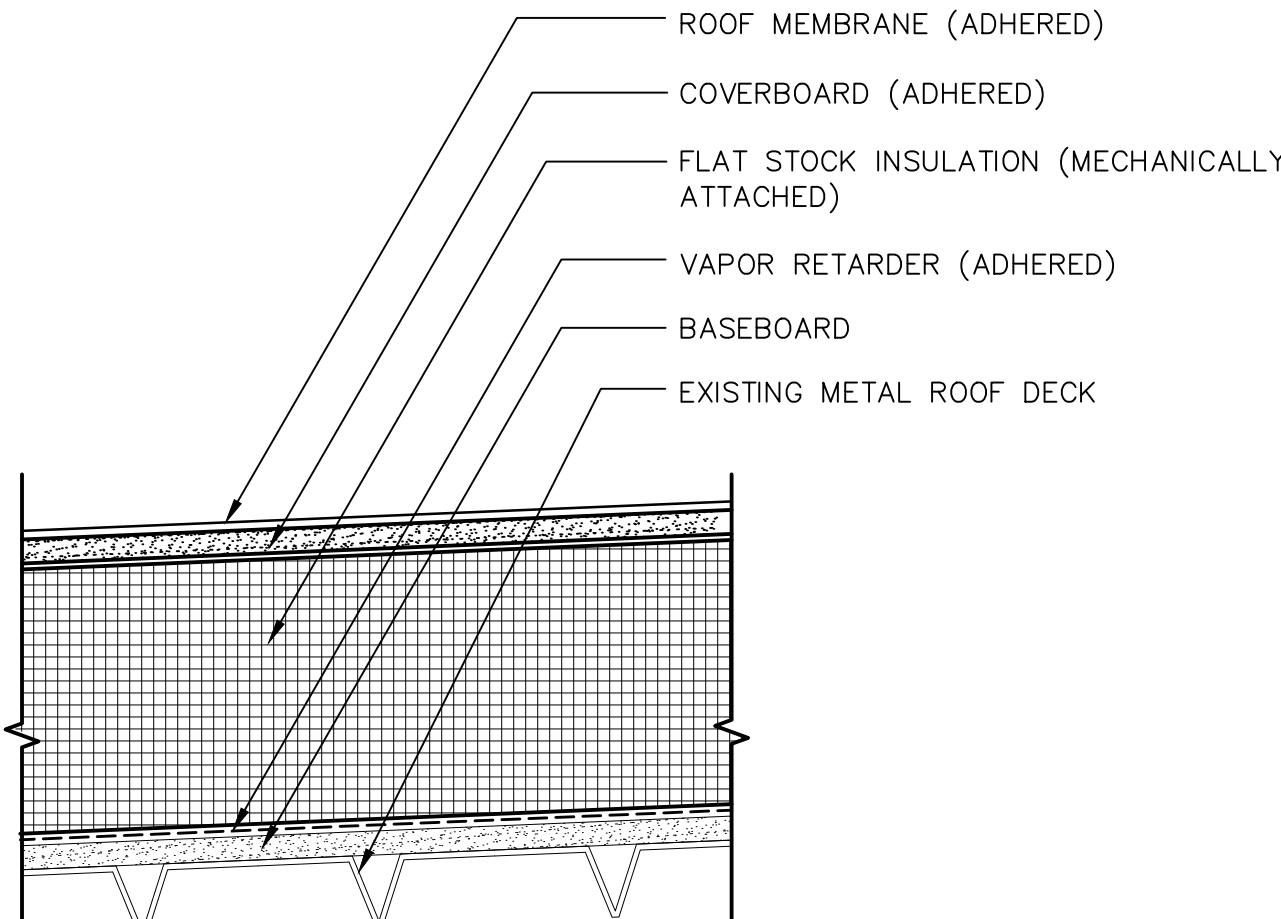
ALL ITEMS ARE NEW UNLESS NOTED OTHERWISE
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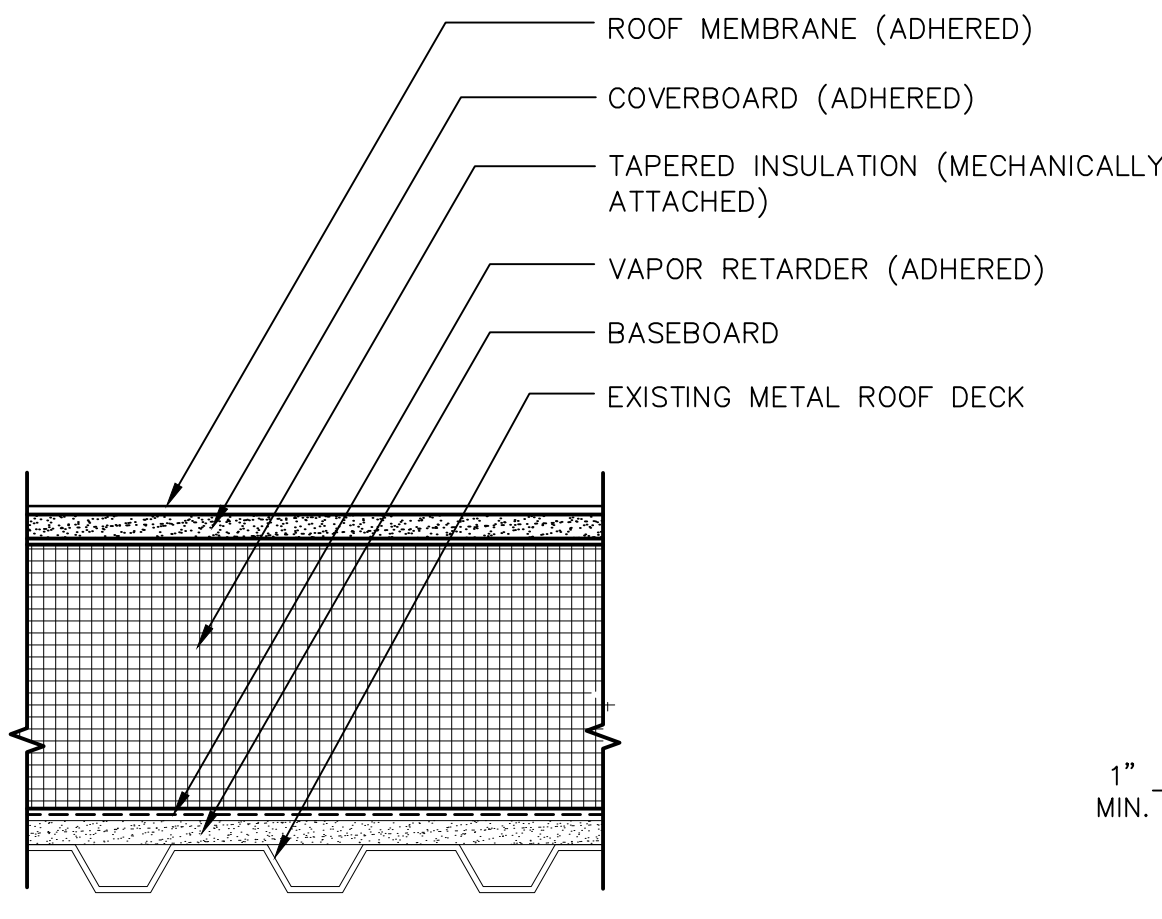
NOTE:
INSULATION ON ROOF AREA L IS FOR
TAPER PURPOSE ONLY AS R-VALUE
IS NOT REQUIRED.



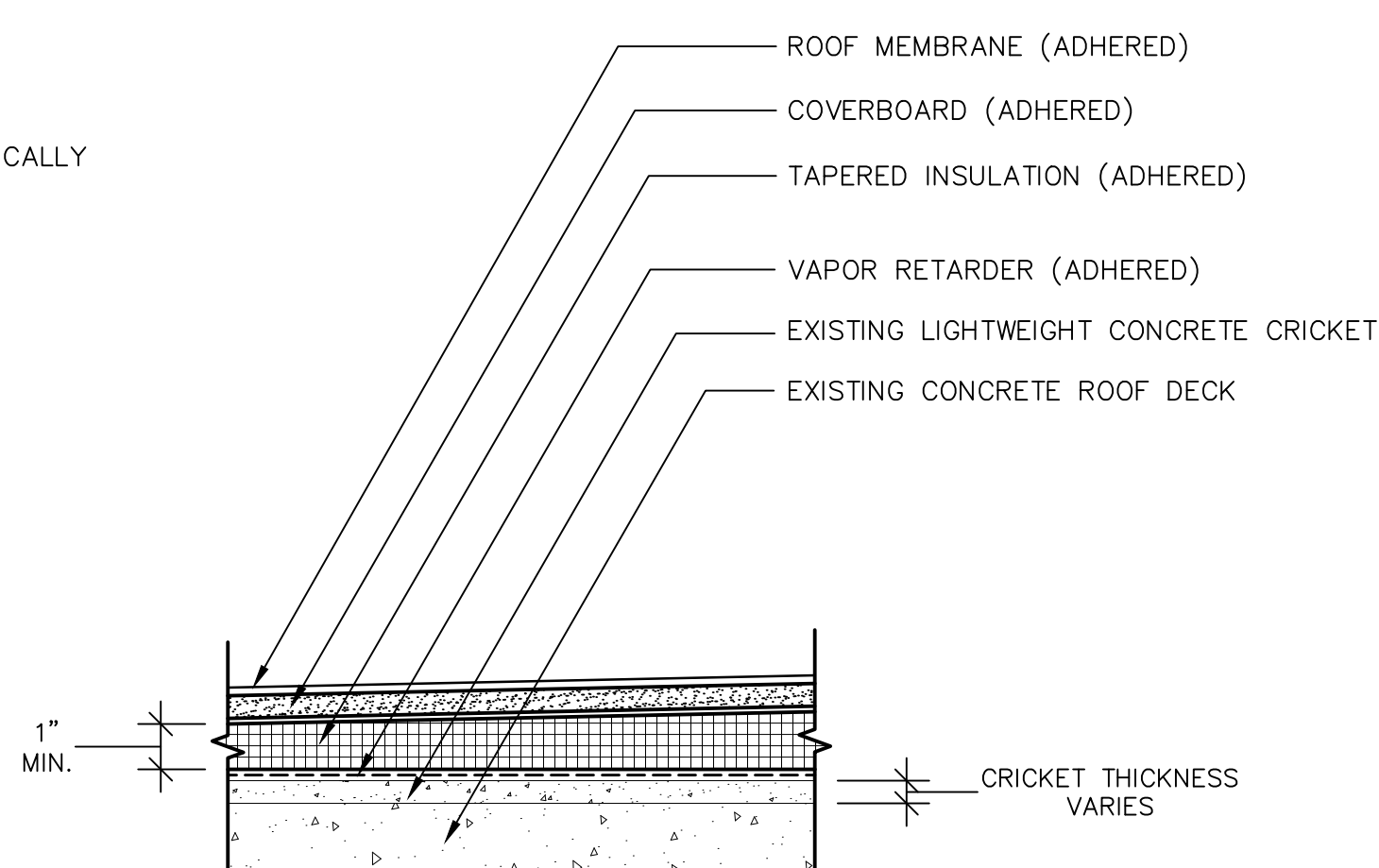
ROOF AREAS A, B, H, J, AND L



ROOF AREA I



ROOF AREAS C, D, E, F, AND K



ROOF AREA G

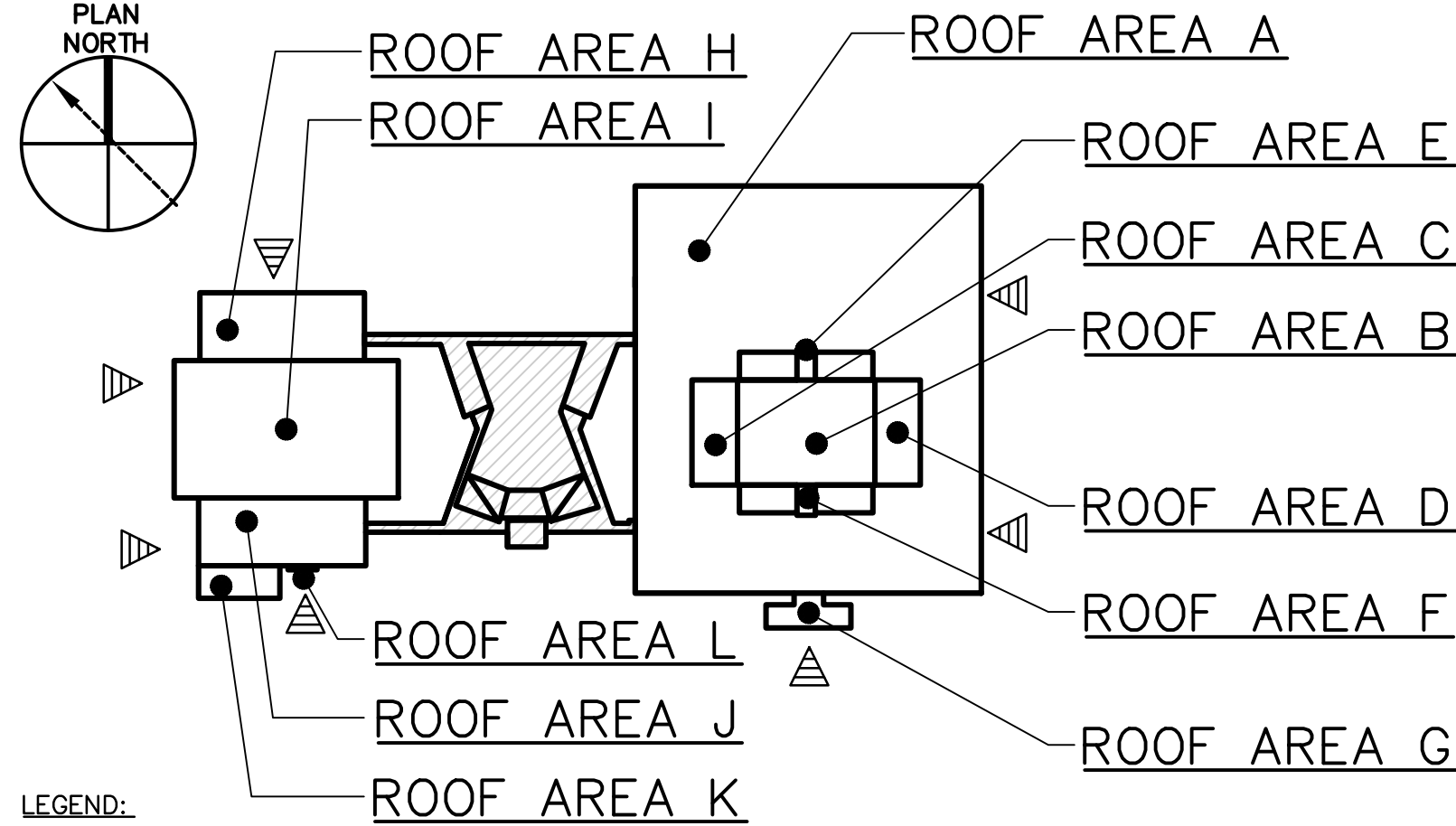
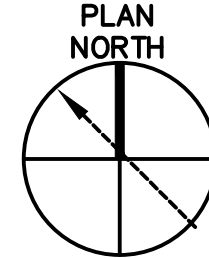
NOTE:
INSULATION ON ROOF AREA G IS
FOR TAPER PURPOSE ONLY AS
R-VALUE IS NOT REQUIRED.

GENERAL NOTES

1. THE INFORMATION SHOWN ON THIS DRAWING HAS BEEN COMPILED FROM VARIOUS SOURCES, AND MAY NOT REFLECT THE ACTUAL CONDITIONS AT THE TIME OF CONSTRUCTION.
2. HATCH PATTERNS ARE FOR REPRESENTATION ONLY AND SHOULD NOT BE USED AS A MEANS OF QUANTIFYING.
3. FOR THE SAKE OF CLARITY, EACH INDIVIDUAL DETAIL ON THE ROOF PLAN HAS NOT BEEN INDICATED.
4. INSTALLATION DETAILS HAVE BEEN INDICATED FOR TYPICAL COMPONENTS AT RANDOM LOCATIONS.
5. THE BUILDING SHALL REMAIN OCCUPIED DURING CONSTRUCTION. THE CONTRACTOR SHALL SCHEDULE AND EXECUTE WORK TO AVOID INTERRUPTIONS TO BUILDING OPERATIONS AND NOTIFY THE OWNER OF WORK LOCATIONS A MINIMUM OF 72 HOURS IN ADVANCE SO THAT OCCUPANCY BELOW THE WORK AREA MAY BE CONTROLLED.
6. REMOVE AND DISPOSE OF THE EXISTING ROOF MEMBRANE AND ALL ASSOCIATED COMPONENTS. THE CONTRACTOR SHALL REPORT DAMP, WET, OR OTHERWISE DETERIORATED COMPONENTS TO THE OWNER PRIOR TO PERFORMING ROOF INSTALLATION WORK.
7. THE CONTRACTOR WILL COORDINATE SHUTDOWN OF AIR INTAKES WITH OWNERS REPRESENTATIVE AS TO PREVENT INTAKE OF FUMES.
8. DISCONNECT, REMOVE, RAISE, REINSTALL, AND RECONNECT ROOFTOP EQUIPMENT AND ASSOCIATED PENETRATIONS TO PROVIDE MINIMUM SPECIFIED FLASHING HEIGHTS.
9. THE CONTRACTOR WILL BE RESPONSIBLE FOR CONFIRMING ALL QUANTITIES AND DIMENSIONS PRIOR TO BIDDING.
10. DISCREPANCIES NOTED BY THE CONTRACTOR IN THE DRAWINGS OR SPECIFICATIONS MUST BE BROUGHT TO THE OWNER'S ATTENTION PRIOR TO BID SUBMISSION.
11. ALL ITEMS OF CONSTRUCTION SHALL BE IN ACCORDANCE WITH LOCAL, STATE, AND FEDERAL REGULATIONS APPLICABLE TO THE PROJECT.
12. THE CONTRACTOR IS CAUTIONED THAT DUE TO BUILDING OCCUPANCY, THE OWNER REQUIRES COMPLIANCE ON WORK HOURS, SCHEDULING, SET-UP, CLEANUP, PARKING, ETC. REFER TO SPECIFICATIONS FOR OWNER REQUIREMENTS.
13. ROOFING AND FLASHING SHALL BE 100% WEATHER TIGHT ON A DAILY BASIS. REMOVE ONLY AS MUCH ROOFING AS CAN BE REPLACED ON THE SAME DAY. PHASED CONSTRUCTION IS NOT PERMITTED WITHOUT PRIOR APPROVAL.
14. DETAILS NOT DEPICTED SHALL BE CONSTRUCTED IN A MANNER CONSISTENT WITH THE DETAIL DRAWINGS.
15. THE CONTRACTOR SHALL BE RESPONSIBLE FOR REMOVING AND LEGALLY DISPOSING OF ALL MATERIALS LISTED AND/OR INDICATED ON THE DRAWINGS OR NOTED IN THE WRITTEN SPECIFICATIONS - TYPICAL.
16. THE BUILDING SHALL BE MAINTAINED IN A DUST FREE CONDITION. THE CONTRACTOR SHALL PROVIDE ALL NECESSARY TEMPORARY PROTECTION TO ENSURE THE INTERIOR OF THE BUILDING REMAINS DUST FREE. NO DEMOLITION CAN PROCEED UNTIL AN APPROVED TEMPORARY PROTECTION SYSTEM IS IN PLACE.
17. THE CONTRACTOR SHALL PROVIDE OVERHEAD PROTECTION AT BUILDING ENTRANCE WAYS IN ACCORDANCE WITH OSHA, LOCAL, STATE, AND FEDERAL GUIDELINES. ALL OVERHEAD PROTECTION SHALL EXTEND A MINIMUM OF 8'-0" BEYOND THE ADJACENT WALL.

KEY PLAN

NOT TO SCALE



- LEGEND:
- ROOF AREA NOT SHOWN ON THIS PLAN
 - BUILDING ENTRANCE



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AMHERST, MA 01002

OWNER
AMHERST-PELHAM REGIONAL SCHOOL DISTRICT
170 CHESNUT STREET
AMHERST, MA 01002

NO.	DATE	DESCRIPTION	BY
PROJECT NO.	833180		
CADD FILE	833180 G101		
DESIGNED BY	CTM		
DRAWN BY	RRF		
CHECKED BY	CM		
DATE	12/08/2017		
DRAWING SCALE	AS NOTED		

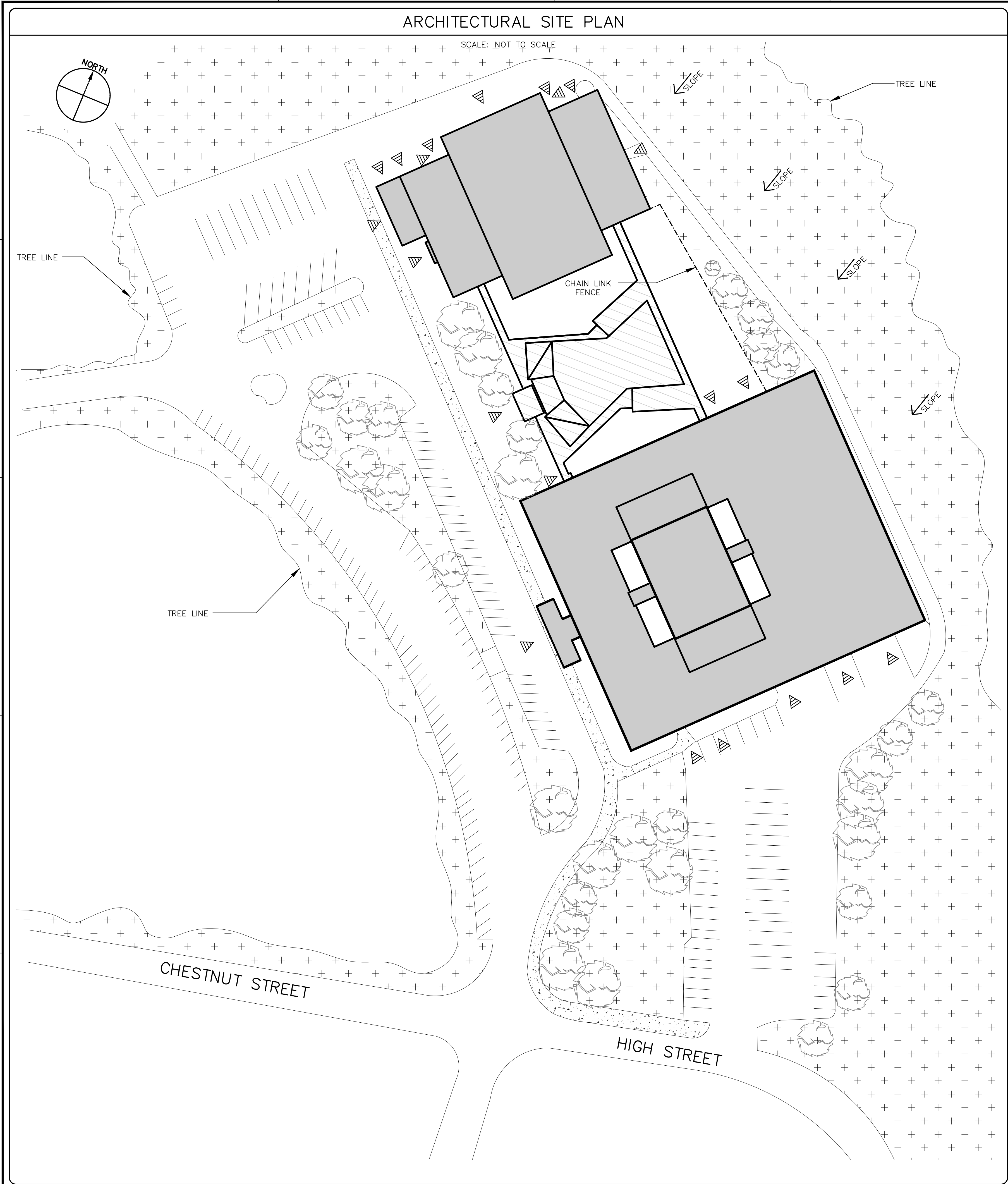
GRAPHIC SCALE

SHEET TITLE

EXISTING AND
NEW ROOF
CROSS SECTIONS,
AND GENERAL
NOTES

DRAWING NO.

G101



SITE LEGEND

- ROOF AREA NOT IN CONTRACT (N.I.C.)
- ROOF AREAS TO BE REMOVED AND REPLACED
- INDICATES GRASS AREA
- SIDEWALK
- INDICATES TREES; SIZE AND HEIGHT VARY
- APPROXIMATE BUILDING PERIMETER
- CHAIN LINK FENCE
- BUILDING ENTRANCE

SITE NOTES

- THE INFORMATION SHOWN ON THIS DRAWING HAS BEEN COMPILED FROM VARIOUS SOURCES, AND MAY NOT REFLECT THE ACTUAL CONDITIONS AT THE TIME OF CONSTRUCTION.
- HATCH PATTERNS ARE FOR REPRESENTATION ONLY AND SHOULD NOT BE USED AS A MEANS FOR QUANTIFYING.
- THE BUILDING SHALL REMAIN OCCUPIED DURING CONSTRUCTION. THE CONTRACTOR SHALL SCHEDULE AND EXECUTE WORK TO AVOID INTERRUPTIONS TO BUILDING OPERATIONS AND NOTIFY THE OWNER OF WORK LOCATIONS A MINIMUM OF 72 HOURS IN ADVANCE IN ORDER THAT OCCUPANCY BELOW THE WORK AREA MAY BE CONTROLLED.
- THE CONTRACTOR IS CAUTIONED THAT DUE TO BUILDING OCCUPANCY, THE OWNER REQUIRES COMPLIANCE ON WORK HOURS, SCHEDULING, SET UP, CLEANUP, PARKING, SECURITY, ETC. REFER TO SPECIFICATIONS FOR THE OWNER'S REQUIREMENTS. REFER TO THE SPECIFICATIONS FOR ADDITIONAL INFORMATION.
- DUE TO THE CONDITIONS AROUND THE PROJECT SITE, THE CONTRACTOR'S STAGING AND STORAGE LOCATIONS WILL BE LIMITED. THE CONTRACTOR MUST COORDINATE AREAS FOR SETUP AND STORAGE OF MATERIALS WITH THE OWNER PRIOR TO STARTING THE PROJECT.
- PROVIDE SITE FENCING AND PLYWOOD AROUND SCAFFOLDING AND STAIR ACCESS TOWERS INSTALLED AT THE STREET LEVEL, AND SHALL BE LOCKED DAILY TO PREVENT UNAUTHORIZED ACCESS.

GALE

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AMHERST, MA 01002

OWNER

AMHERST-PELHAM REGIONAL SCHOOL DISTRICT
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AMHERST, MA 01002

NO.	DATE	DESCRIPTION	BY	
PROJECT NO.	833180			
CADD FILE	833180 G101			
DESIGNED BY	CTM			
DRAWN BY	RRF			
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DATE	12/08/2017			
DRAWING SCALE	AS NOTED			

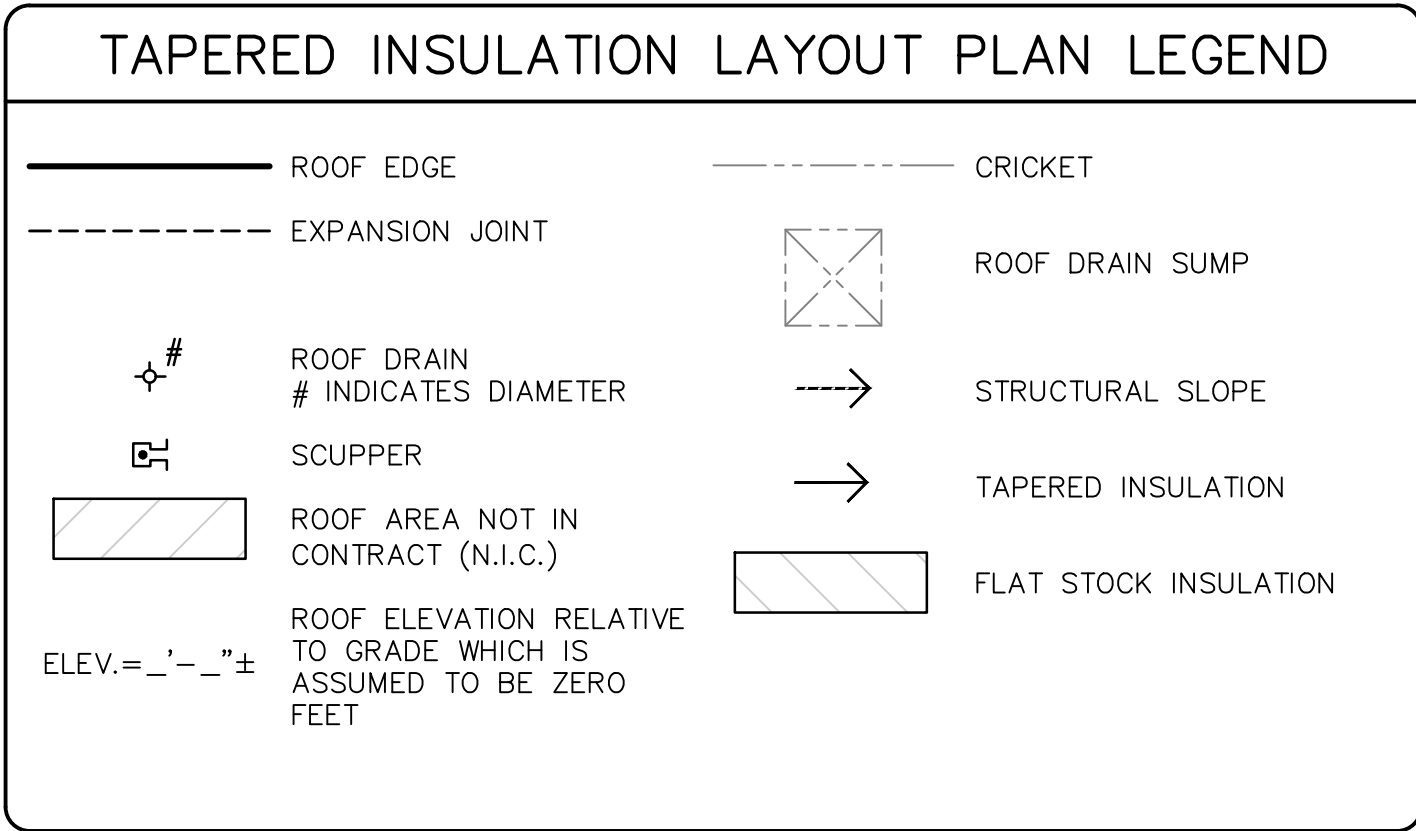
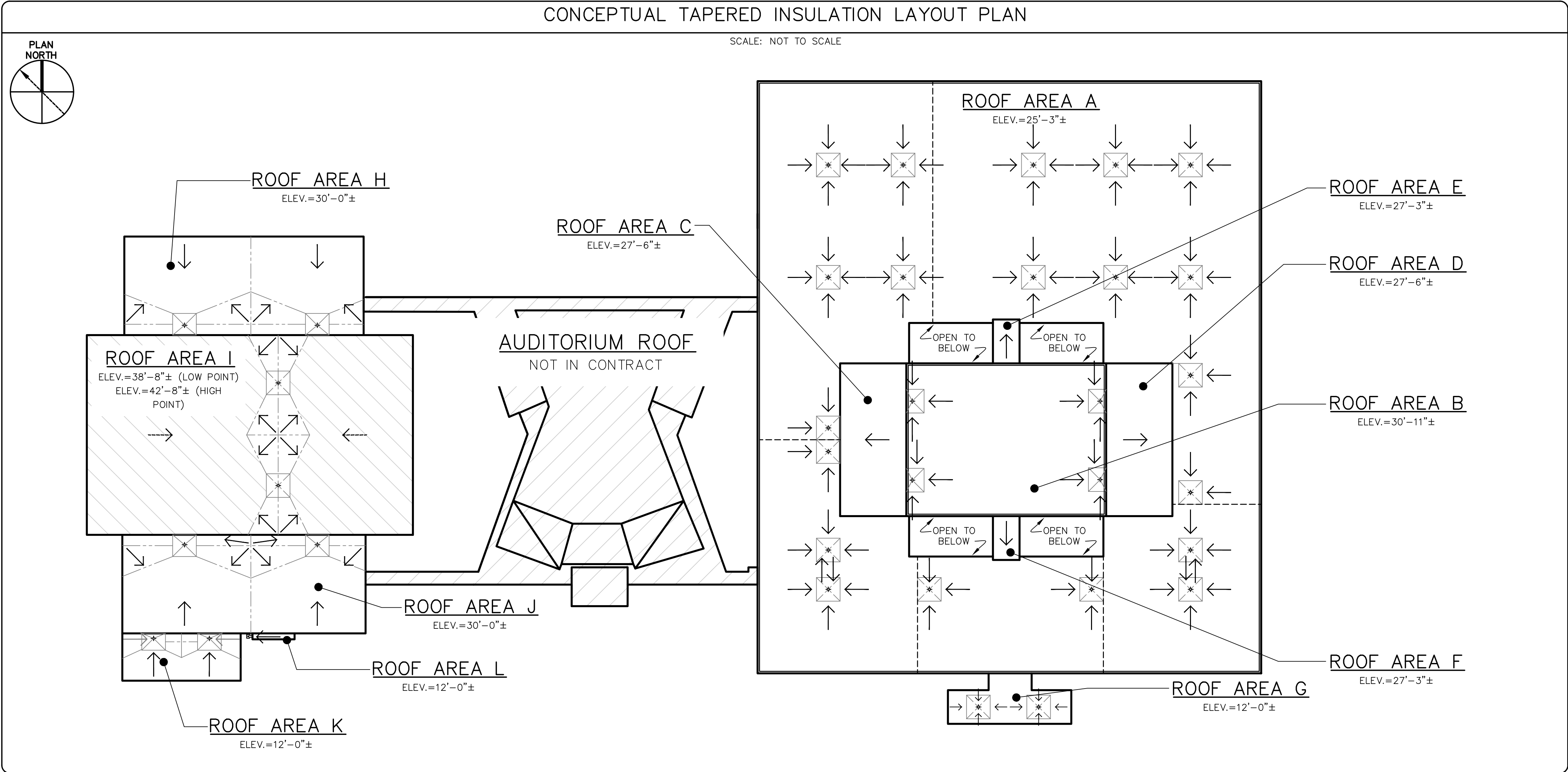
GRAPHIC SCALE

SHEET TITLE

SITE PLAN AND
NOTES

DRAWING NO.

G102



- TAPERED INSULATION LAYOUT PLAN NOTES
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 2. HATCH PATTERNS ARE FOR REPRESENTATION ONLY AND SHOULD NOT BE USED AS A MEANS FOR QUANTIFYING.
 3. CONCEPTUAL TAPERED INSULATION LAYOUT TO PROVIDE MINIMUM INSULATION HEIGHTS AT ROOF TO WALL AND PERIMETER LOCATIONS. SPECIFIC TAPERED LAYOUT TO BE DEVELOPED BY THE CONTRACTOR IN CONJUNCTION WITH THE INSULATION SUPPLIER. ALL ROOF PENETRATIONS ARE NOT SHOWN ON THIS CONCEPTUAL LAYOUT; PROVIDE CRICKETS BEHIND UNITS TO PROMOTE DRAINAGE. CONTRACTOR TO VERIFY EXISTING ROOF DECK SLOPE AS THE WORK PROGRESSES.
 4. ADJUSTMENTS TO WOOD BLOCKING HEIGHTS AND CONFIGURATIONS OF FLASHINGS TO CONFORM TO THE TAPERED INSULATION LAYOUT IS THE RESPONSIBILITY OF THE CONTRACTOR. NO ADDITIONAL COSTS WILL BE ALLOWED TO THE CONTRACTOR IF PROPOSED DETAIL CONFIGURATIONS ARE ALTERED.
 5. ANY PONDING WATER ENCOUNTERED AFTER THE INSTALLATION OF THE NEW ROOF SYSTEM WILL BE ADDRESSED BY THE CONTRACTOR AT NO ADDITIONAL COST TO THE OWNER. PONDING WATER IS DEFINED AS 1/2" OR GREATER OF STANDING WATER WITHIN A 100 SQUARE FOOT AREA AFTER A 24 HOUR PERIOD.
 6. REVIEW OF THE EXPOSED ROOF DECKS BY THE CONTRACTOR WILL BE REQUIRED PRIOR TO PLACEMENT OF THE INSULATION SYSTEM TO DETERMINE POTENTIAL UNFORESEEN LOW AREAS.
 7. COORDINATE FINAL WOOD BLOCKING HEIGHTS WITH THAT OF THE FINAL TAPERED INSULATION HEIGHTS. WOOD BLOCKING HEIGHTS SHALL REMAIN CONSTANT AROUND EACH INDIVIDUAL ROOF AREA. REFER TO THE CONSTRUCTION DETAILS FOR ADDITIONAL INFORMATION. INSTALL TAPERED EDGE STRIPS AT ROOF EDGES TO PROVIDE A SMOOTH TRANSITION.
 8. CONTRACTOR IS TO PROVIDE TAPERED INSULATION OVERLAY AT RIDGE LOCATION, SO THAT THE HEIGHT OF THE INSULATION MATCHES ON BOTH SIDES OF THE RIDGE, AND PROVIDES A SMOOTH TRANSITION.



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PROJECT

ROOF REPLACEMENT
AND ASSOCIATED WORK AT THE
AMHERST REGIONAL MIDDLE SCHOOL
AMHERST, MA 01002

OWNER

AMHERST-PELHAM REGIONAL SCHOOL DISTRICT
170 CHESNUT STREET
AMHERST, MA 01002

NO.	DATE	DESCRIPTION	BY
PROJECT NO.	833180		
CADD FILE	833180 G101		
DESIGNED BY	CTM		
DRAWN BY	RRF		
CHECKED BY	CM		
DATE	12/08/2017		
DRAWING SCALE	AS NOTED		

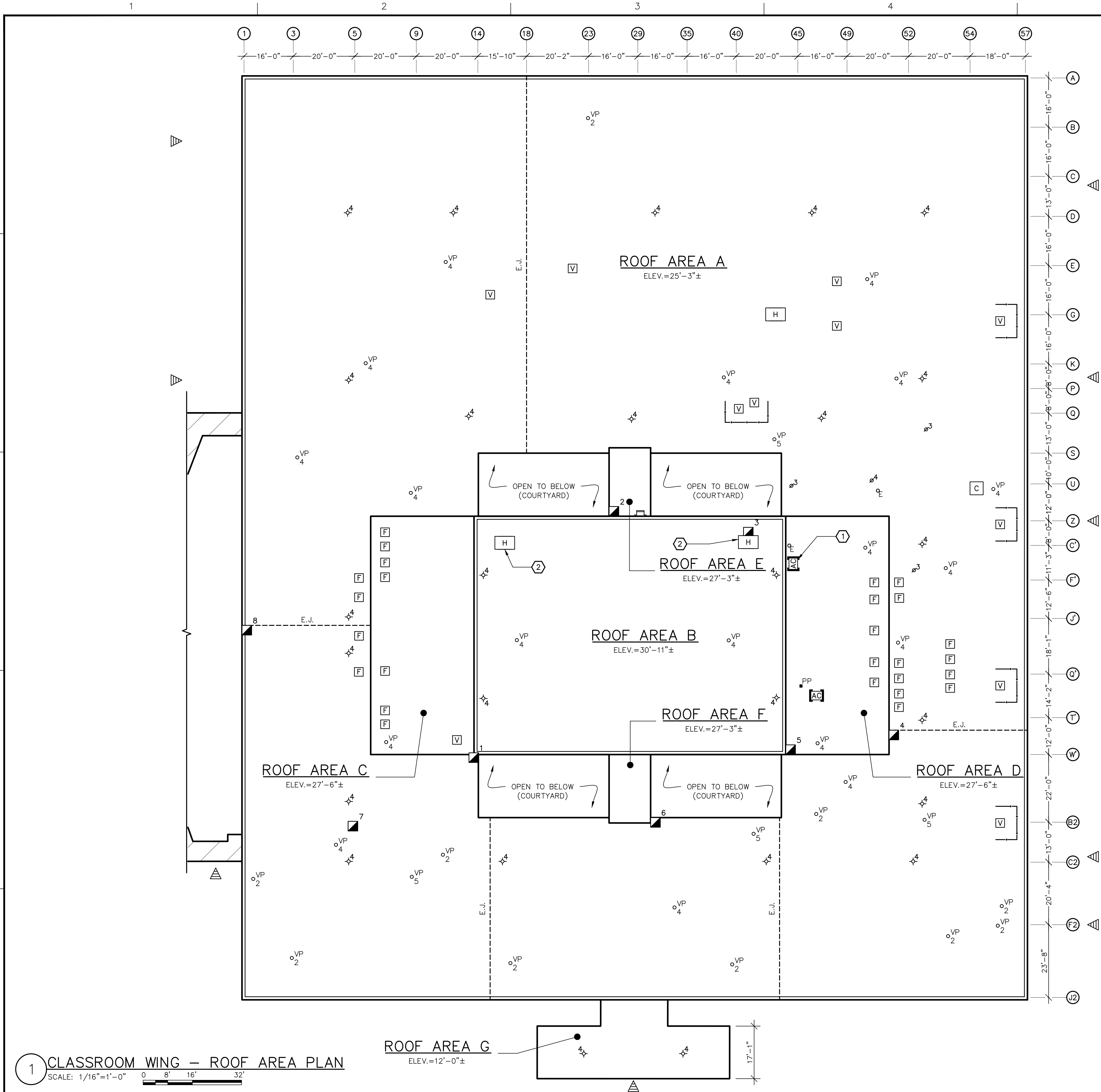
GRAPHIC SCALE

SHEET TITLE

CONCEPTUAL
TAPERED
INSULATION PLAN
AND NOTES

DRAWING NO.

G103



NOTE: RE-SECURE THE EXISTING METAL DECK AT ALL PERIMETER AND CORNER LOCATIONS ON ROOF AREAS A AND B.

KEY NOTES

- 1 DISCONNECT AC UNIT AND PLACE ON SLEEPERS.
- 2 REMOVE ROOF HATCHES AND INFILL WITH METAL DECK.
- 3 NOT USED ON THIS SHEET
- 4 NOT USED ON THIS SHEET
- 5 NOT USED ON THIS SHEET
- 6 NOT USED ON THIS SHEET
- 7 NOT USED ON THIS SHEET

GENERAL NOTES

- THE INFORMATION SHOWN ON THIS DRAWING HAS BEEN COMPILED FROM VARIOUS SOURCES, AND MAY NOT REFLECT THE ACTUAL CONDITIONS AT THE TIME OF CONSTRUCTION.
- FOR THE SAKE OF CLARITY, EACH INDIVIDUAL DETAIL ON THE ROOF PLAN HAS NOT BEEN INDICATED. INSTALLATION DETAILS HAVE BEEN INDICATED FOR TYPICAL COMPONENTS AT RANDOM LOCATIONS.
- HATCH PATTERNS ARE FOR REPRESENTATION ONLY AND SHOULD NOT BE USED AS MEANS FOR QUANTIFYING.
- COLUMN LINE DIMENSIONS ARE BASED ON THE ORIGINAL BUILDING DOCUMENTS.

LEGEND

ALL ITEMS ARE EXISTING UNLESS DESIGNATED OTHERWISE

—	ROOF EDGE	PP	POURABLE SEALER POCKET
—	PARAPET	#	ROOF DRAIN # INDICATES DIAMETER (INCH)
E.J.	EXPANSION JOINT	#	PIPE PENETRATION # INDICATES DIAMETER (INCH)
—	PROPOSED GUARDRAIL	oVP	VENT PIPE # INDICATES DIAMETER (INCH)
M	MECHANICAL PIPING	⊕	ELECTRICAL PENETRATION
---	CRICKET	→	STRUCTURAL SLOPE
C	CHIMNEY	V	VENT
⊕	ELECTRICAL PENETRATION	AC	AIR CONDITIONING UNIT ON SLEEPERS
△	BUILDING ENTRANCE		ROOF AREA NOT IN CONTRACT
F	FAN		ROOF ELEVATION RELATIVE TO GRADE WHICH IS ASSUMED TO BE ZERO FEET
oVP	VENT PIPE	⊕	COLUMN LINE
H	ACCESS HATCH		
L	ACCESS LADDER		
■	TEST CUT LOCATION		

KEY PLAN

NOT TO SCALE

PLAN NORTH

LEGEND:
■ ROOF AREAS SHOWN ON THIS PLAN
▨ ROOF AREA NOT IN CONTRACT

1 CLASSROOM WING — ROOF AREA PLAN
SCALE: 1/16"=1'-0" 0 8' 16' 32'



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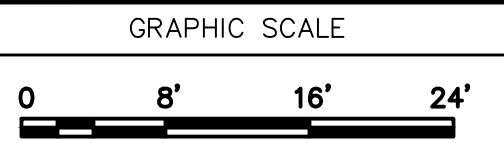
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PROJECT NO.	833180		
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DRAWN BY	RRF		
CHECKED BY	CM		
DATE	12/08/2017		
DRAWING SCALE	1/16"=1'-0"		



SHEET TITLE

CLASSROOM WING
ROOF AREA PLAN

DRAWING NO.
A101

1

2

3

4

5

6

E

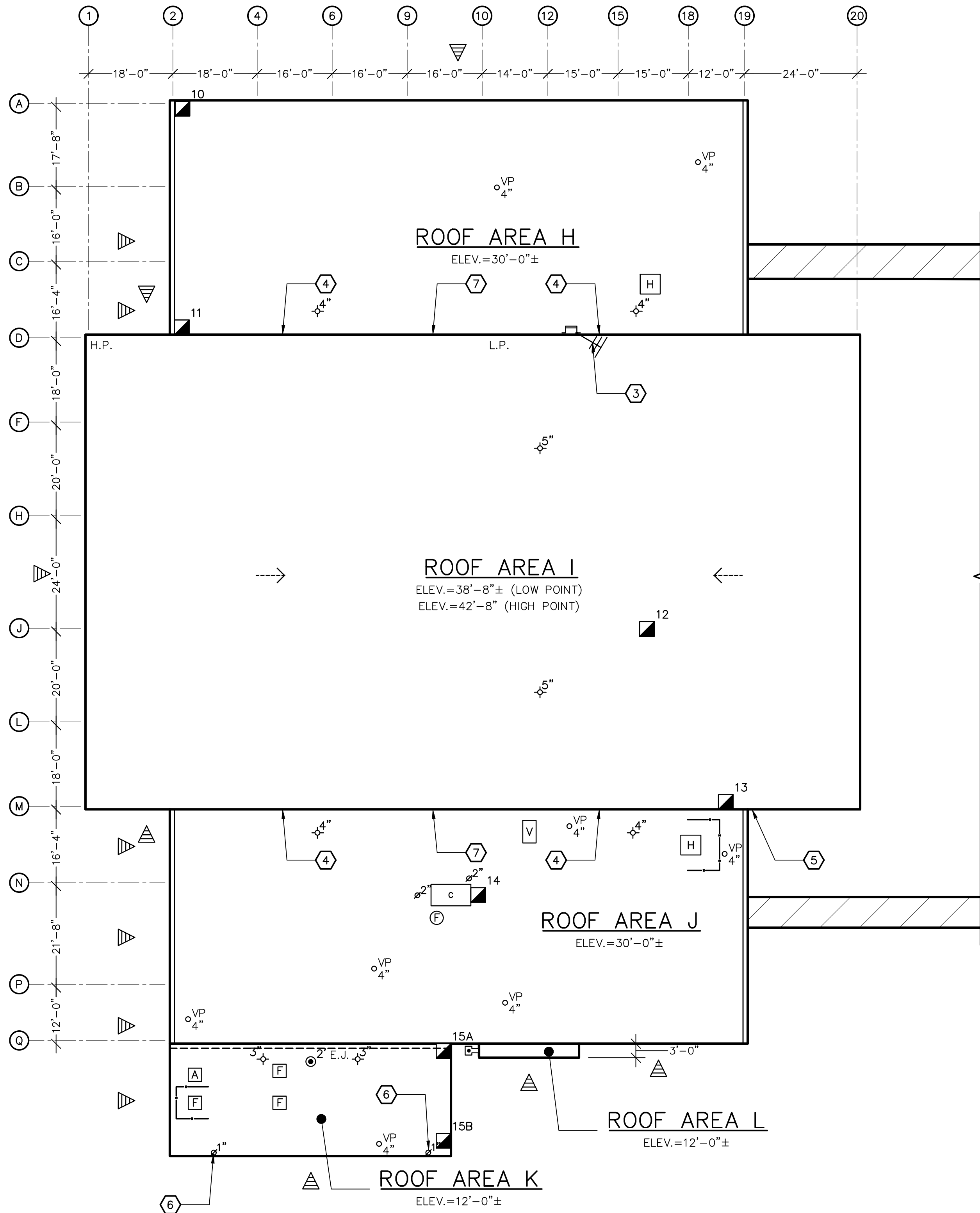
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C

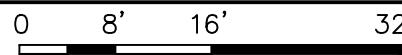
B

A

NOTE: RE-SECURE THE EXISTING METAL DECK AT ALL PERIMETER AND CORNER LOCATIONS ON ROOF AREAS H, I, AND J.



1 GYM/POOL ROOF AREA PLAN
SCALE: 1/16"=1'-0"



KEY NOTES

- 1 NOT USED ON THIS SHEET
- 2 NOT USED ON THIS SHEET
- 3 ANTENNA/WEATHER EQUIPMENT MOUNTED ON LADDER.
- 4 MISSING SEALANT IN MASONRY CONTROL JOINT TO BE REPAIRED (HEIGHTS VARY APPROXIMATELY 9' TO 12')
- 5 LARGE STEP CRACK IN BRICK MASONRY RISING WALL TO BE REBUILT AND EXPANSION JOINT TO BE SAWCUT.
- 6 GUY WIRE ANCHOR FOR SMOKE STACK TO BE RAISED
- 7 REMOVE AND REPLACE PERIMETER SEALANT AROUND LOUVER (APPROX. 30 LINEAR FEET)

GENERAL NOTES

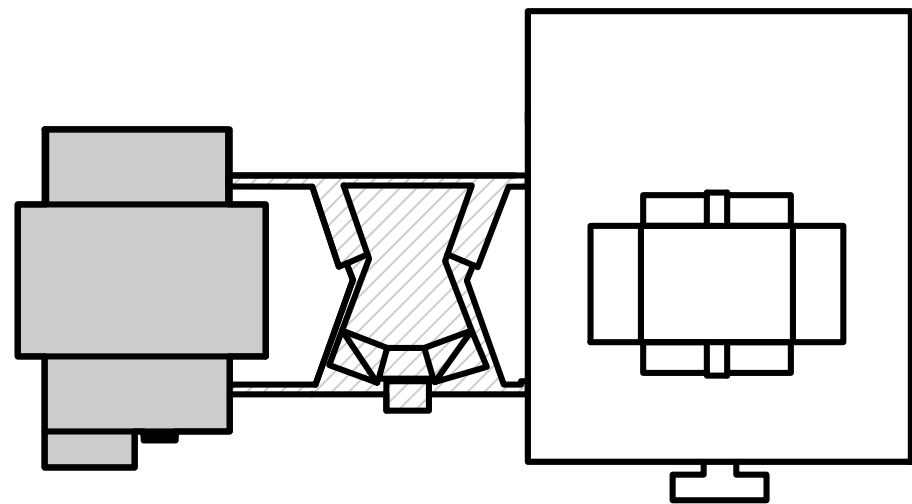
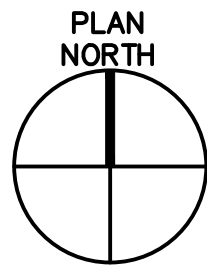
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3. HATCH PATTERNS ARE FOR REPRESENTATION ONLY AND SHOULD NOT BE USED AS MEANS FOR QUANTIFYING.
4. COLUMN LINE DIMENSIONS ARE BASED ON THE ORIGINAL BUILDING DOCUMENTS.

LEGEND

ALL ITEMS ARE EXISTING UNLESS DESIGNATED OTHERWISE			
—	ROOF EDGE	⊘	ROOF DRAIN
- - - E.J. - - -	EXPANSION JOINT	#	# INDICATES DIAMETER (INCH)
==	PARAPET	---	STRUCTURAL SLOPE
—	GUARDRAIL	●	STACK
[C]	CHIMNEY	V	VENT
(F)	FAN	[Hatched]	ROOF AREA NOT IN CONTRACT
⊘ VP	VENT PIPE		ROOF ELEVATION RELATIVE TO GRADE WHICH IS ASSUMED TO BE ZERO FEET
#	# INDICATES DIAMETER (INCH)	ELEV. = _'-_ " ±	
[H]	ACCESS HATCH	[S]	SCUPPER WITH DOWNSPOUT
[L]	ACCESS LADDER	[A]	ANTENNA/WEATHER EQUIPMENT
⊘	PIPE PENETRATION	[#]	COLUMN LINE
#	# INDICATES DIAMETER (INCH)	H.P.	HIGH POINT
[A]	TEST CUT LOCATION	L.P.	LOW POINT
[]	ABANDONED CURB		
[]	BUILDING ENTRANCE		

KEY PLAN

NOT TO SCALE



- LEGEND:
- [Solid Grey] ROOF AREAS SHOWN ON THIS PLAN
 - [Hatched] ROOF AREA NOT IN CONTRACT



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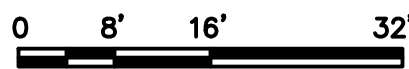
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DATE	12/08/2017		
DRAWING SCALE	1/16"=1'-0"		

GRAPHIC SCALE



SHEET TITLE

GYM/POOL
ROOF AREA PLAN

DRAWING NO.

A102